Advanced Cognitive Neuroscience

Week 45: What about that other cortex? - the cerebellar one

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The course plan

Week 37:
Monday: Workshop paradigm: Measuring visual subjective experience
MR Recordings (Lau)

Week 38:
Monday: What is it all about? (Lau)
Workshop 1: MR-recordings (Laura & Lau)

Week 39:
Monday: Basic physiology and Evoked responses (Lau)
Wednesday: Preprocessing and evoked responses (Laura)

Week 40:
Monday: Forward modelling and inverse modelling (Lau)
Wednesday: Source reconstruction (Laura)

Week 41:
Monday: Multivariate statistics (Laura) (Lecture)
Wednesday: Multivariate statistics (Laura) (Coding)
Deadline for feedback: Video Explainer

Week 43:
Monday: Inverse modelling: beamforming (Lau) (Lecture)
Wednesday: Inverse modelling: beamforming (Laura) (Coding)

Week 44:
Monday From single subject analysis to group analysis (Laura)
Wednesday: Work on your reports (Laura)

Week 45:
Monday: What about that other cortex? - the cerebellar one (Lau)
Wednesday: Work on your reports, continued (Laura)

Week 46:
Wednesday: Optically pumped magnetoencephalography (Laura)
Wednesday: Respiration (Laura)
Deadline for feedback: Lab report

Week 47:
No teaching: prepare presentations!

Week 48:
Monday: Oral presentations (part 1) (Lau)
Wednesday: Oral presentations (part 2) (Lau)

Week 49:
Monday: Oral presentations (part 3) (Lau)
Wednesday: Oral presentations (part 4) (Lau)

Week 50:
Wednesday: What was it all about (Lau)

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LAST TIME

Beamforming

&

Creating a glossary of terms for the course so far

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Glossary of terms

- Super-terms to create glossary around
  - Basic physiology
  - Multivariate statistics
  - Forward modelling (and dipole estimation)
  - Inverse modelling; MNE and beamforming

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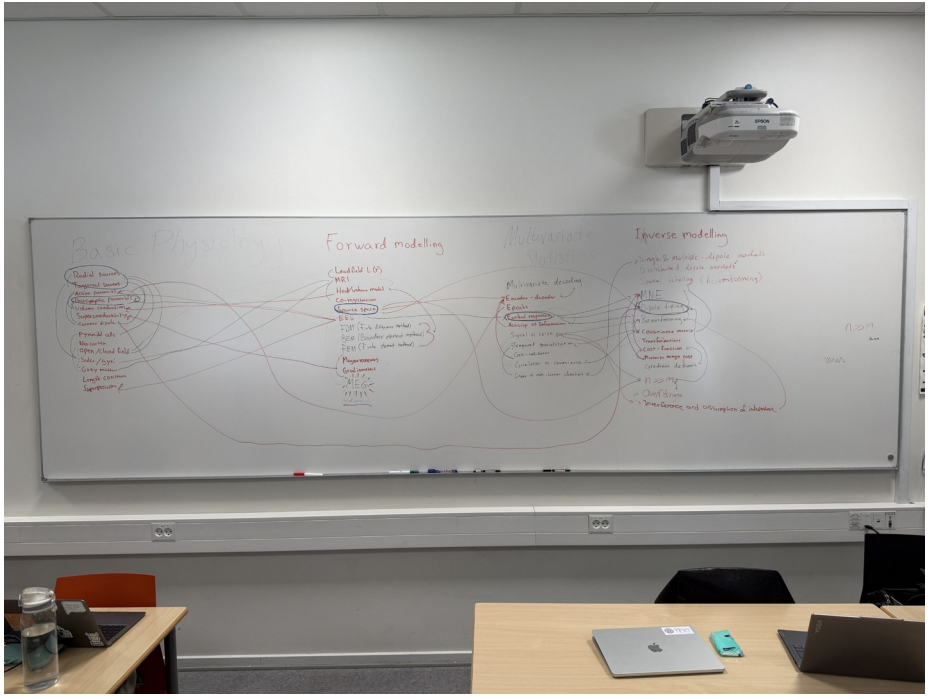
Glossary of terms

PROCEDURE

1) (10 minutes): in your study groups find all the sub-terms you think belong to each of the four super-terms
2) (10 minutes) in pairs of groups, go and write the sub-terms for each term on the whiteboard; start making internal arrows connecting them
3) (10 minutes) start connecting the sub-terms and the super-terms all across the board (mayhem will ensue)
4) (10 minutes) discussing it all together
5) Take pictures to preserve for posterity

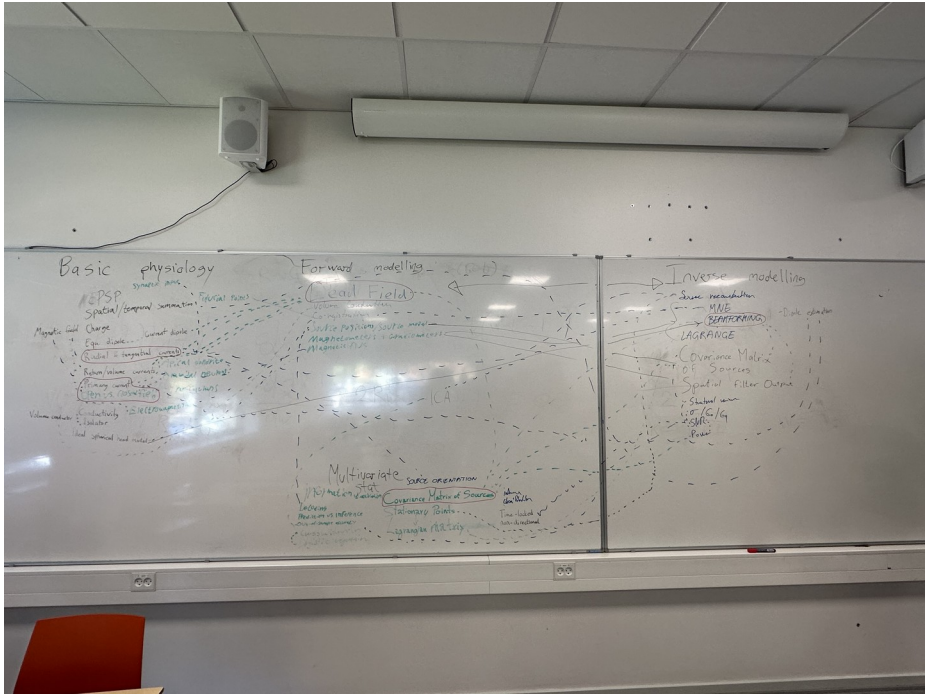
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Glossaries



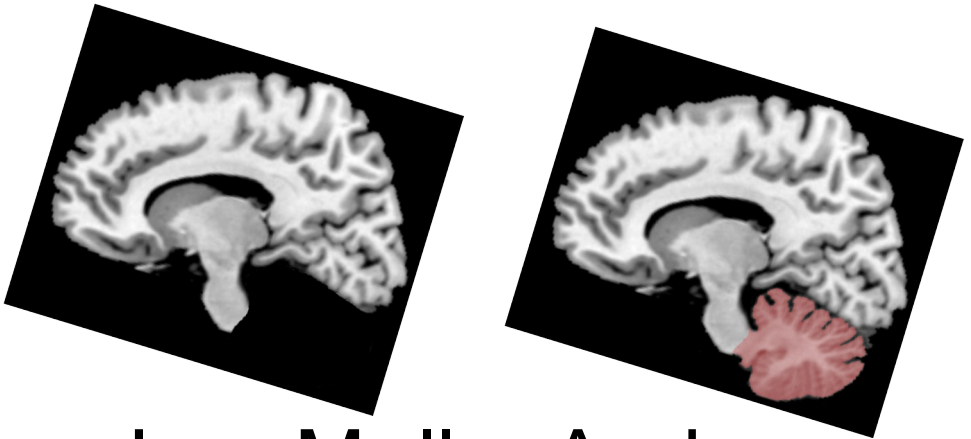
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Glossaries



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Taking the cerebellum seriously



Lau Møller Andersen

Based partly on this talk: Universität Münster, December 12th 2024

AARHUS UNIVERSITY

Department of Linguistics, Cognitive Science and Semiotics

Karolinska Institutet

CFIN

AARHUS UNIVERSITY HOSPITAL

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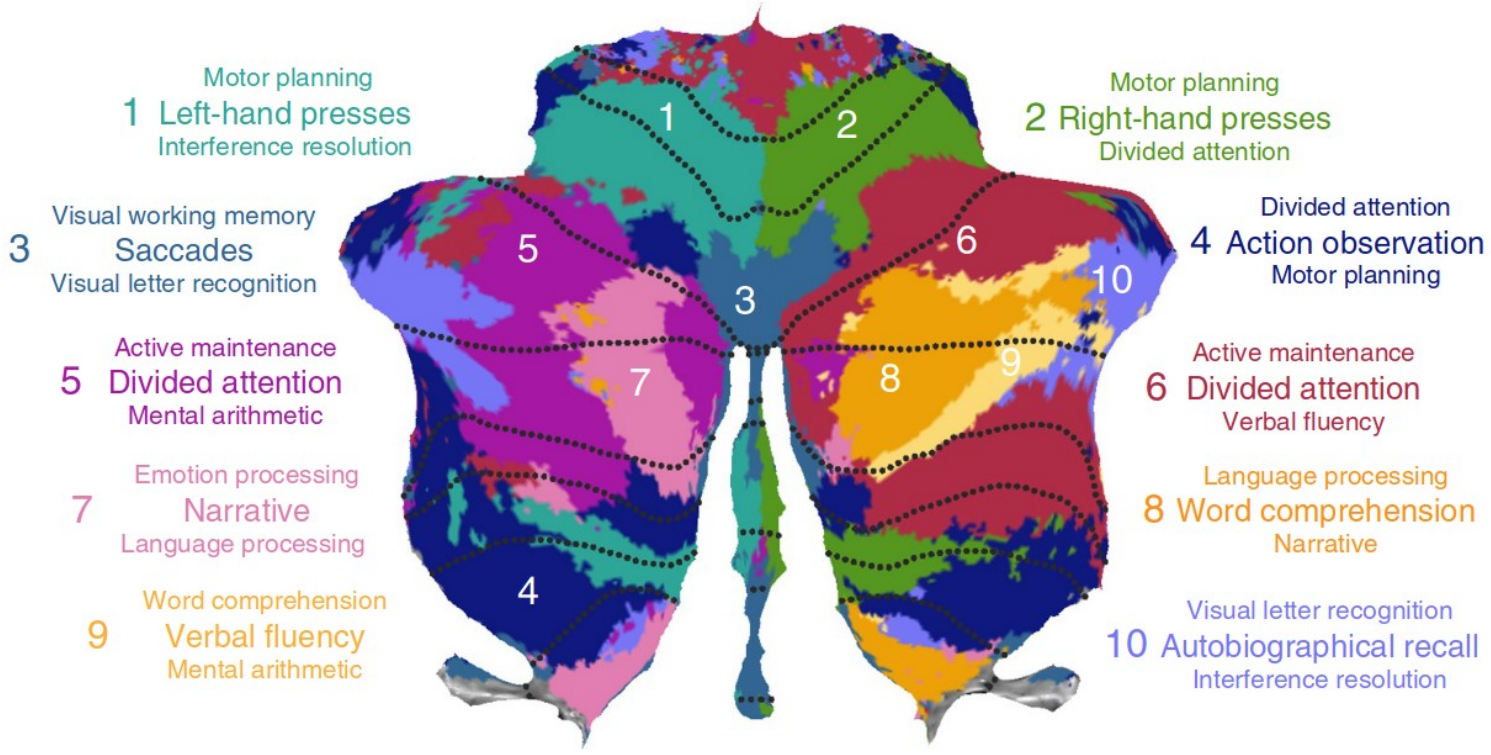
Universität Münster

Key aspects
OBJECTIVES

1) To show that the cerebellum is integrated with the cerebrum, mid-brain and the body
   - How is timing represented in the cerebellum?
   - Cesario et al. 2020: *Your Brain Is Not an Onion With a Tiny Reptile Inside*
2) To show that it is possible to measure its activity reliably in a non-invasive manner
   - AKA: please do not cut the cerebellum from your “whole brain”-analyses

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A primitive part of the brain?

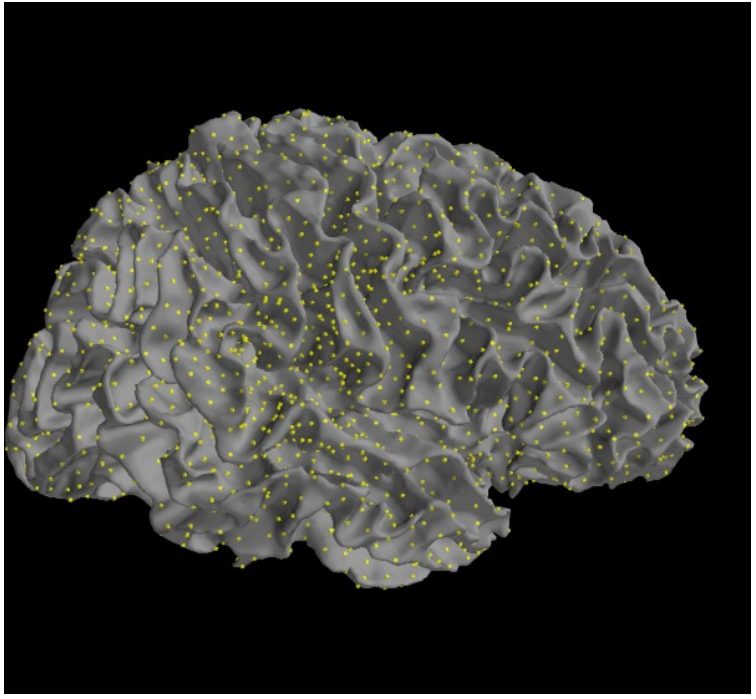


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(King et al. 2019)

Impossible to measure?

OR NOT MODELLED AT ALL?



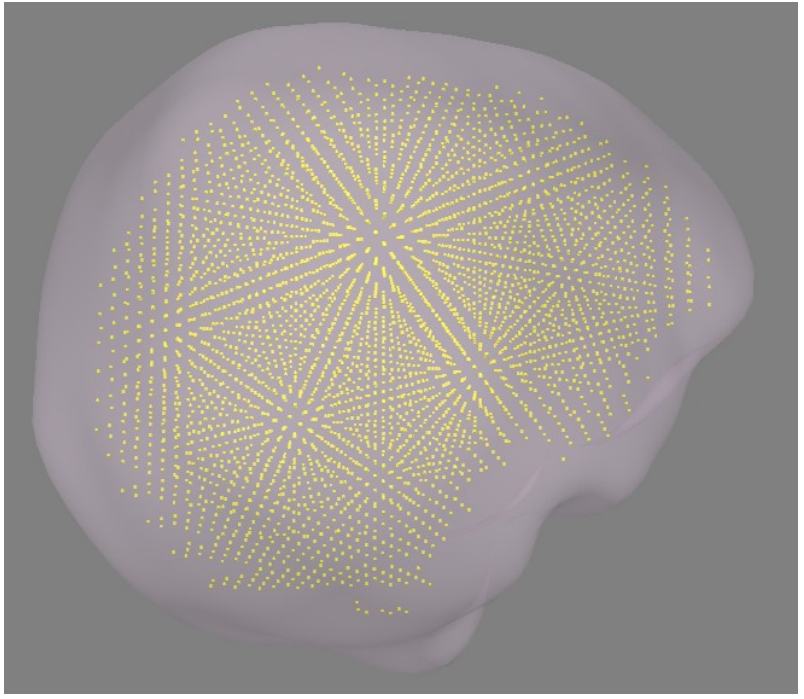
Restricted to the cortical surface

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https://github.com/ualsbombe/2025_advanced_cognitive_neuroscience/blob/main/experiment/mr_processing/python/bem_processing.py

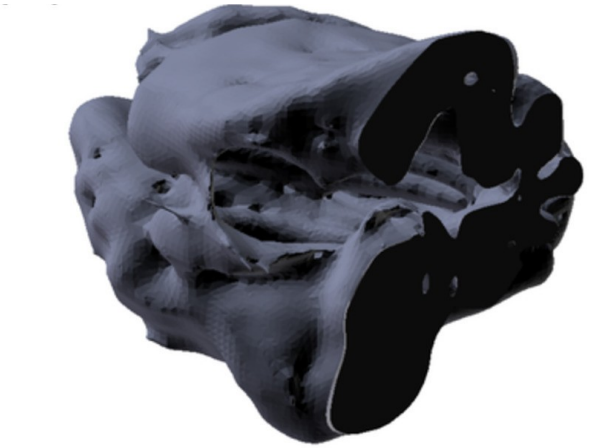
Source model – volumetric

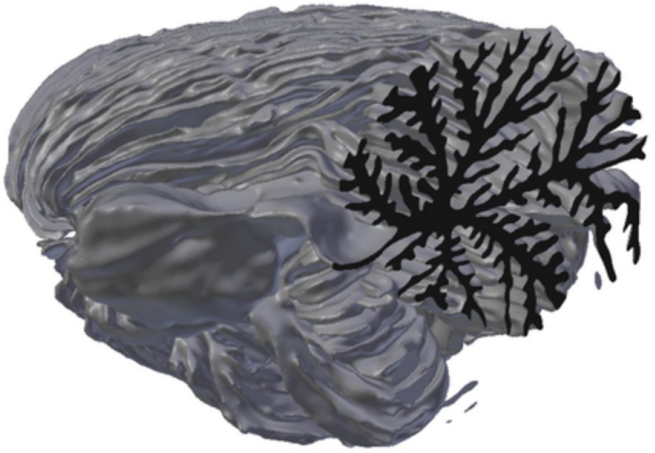
COULD WE RESTRICT IT?



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Restricting cerebellar sources to the surface?



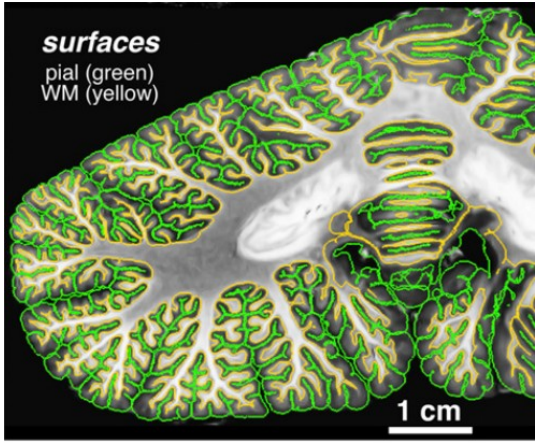


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Samuelsson et al. 2020

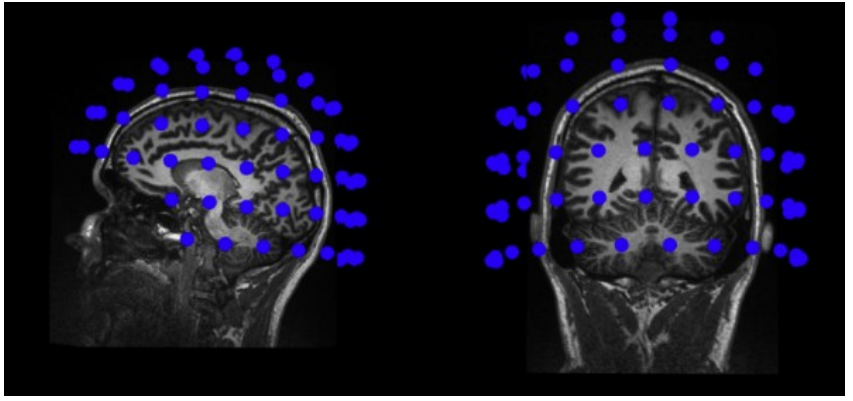
Cerebellar challenges

The cerebellar cortex is **highly convoluted** (78% of the surface of the cerebral cortex, which may lead to signal cancellation).



Sereno et al. 2020

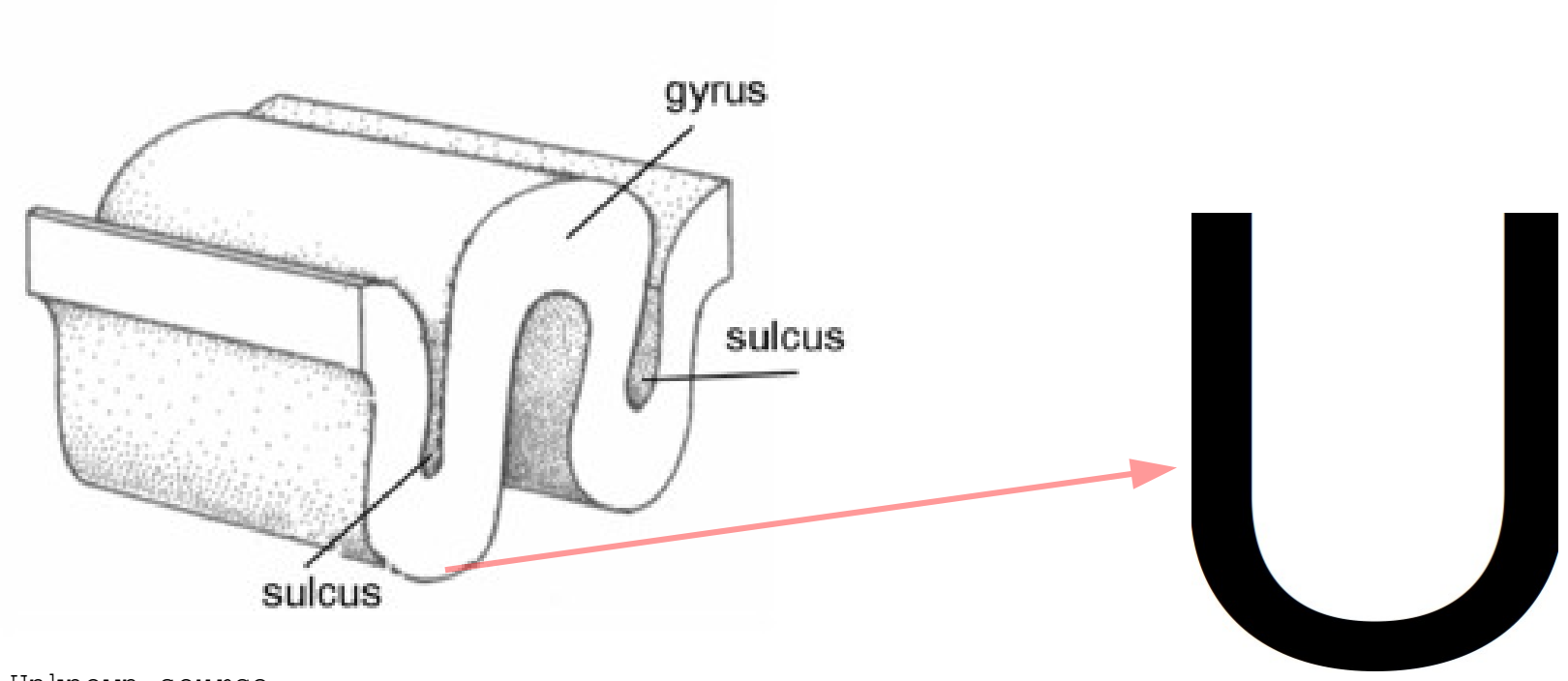
The cerebellar cortex is at a **remote location** which leads to mixing at the recording sites and low SNR



Andersen, Jerbi & Dalal 2020

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Schematic of summation/cancellation

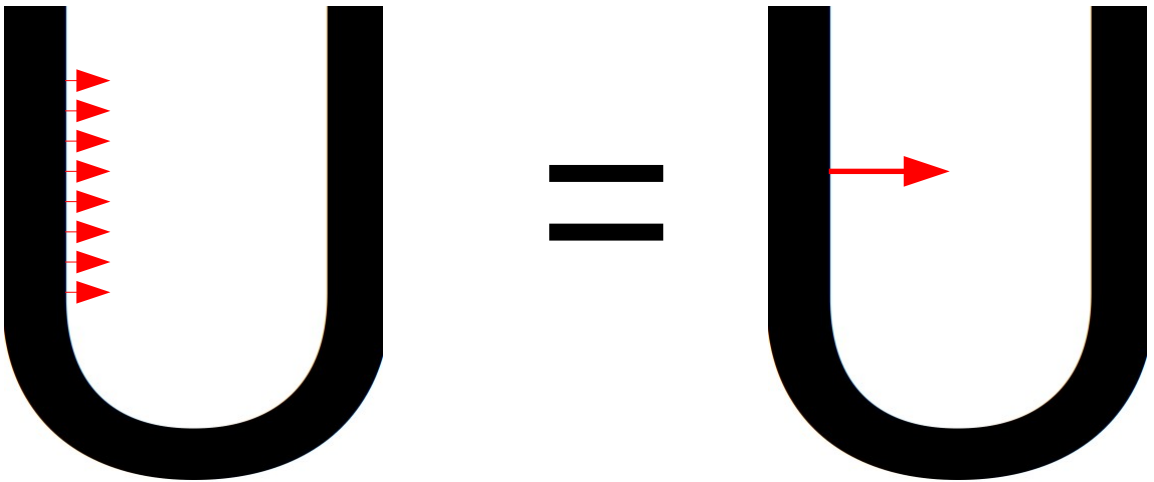


Unknown source

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Schematic of summation/cancellation

Individually considered, sources not measurable at a distance

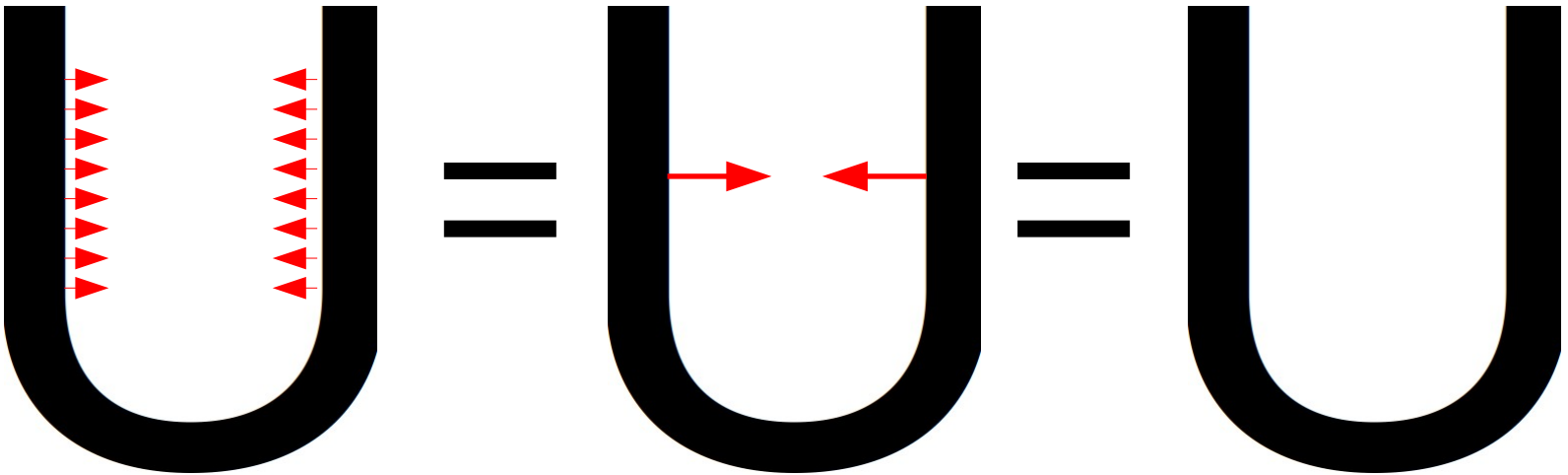


But the sum is...

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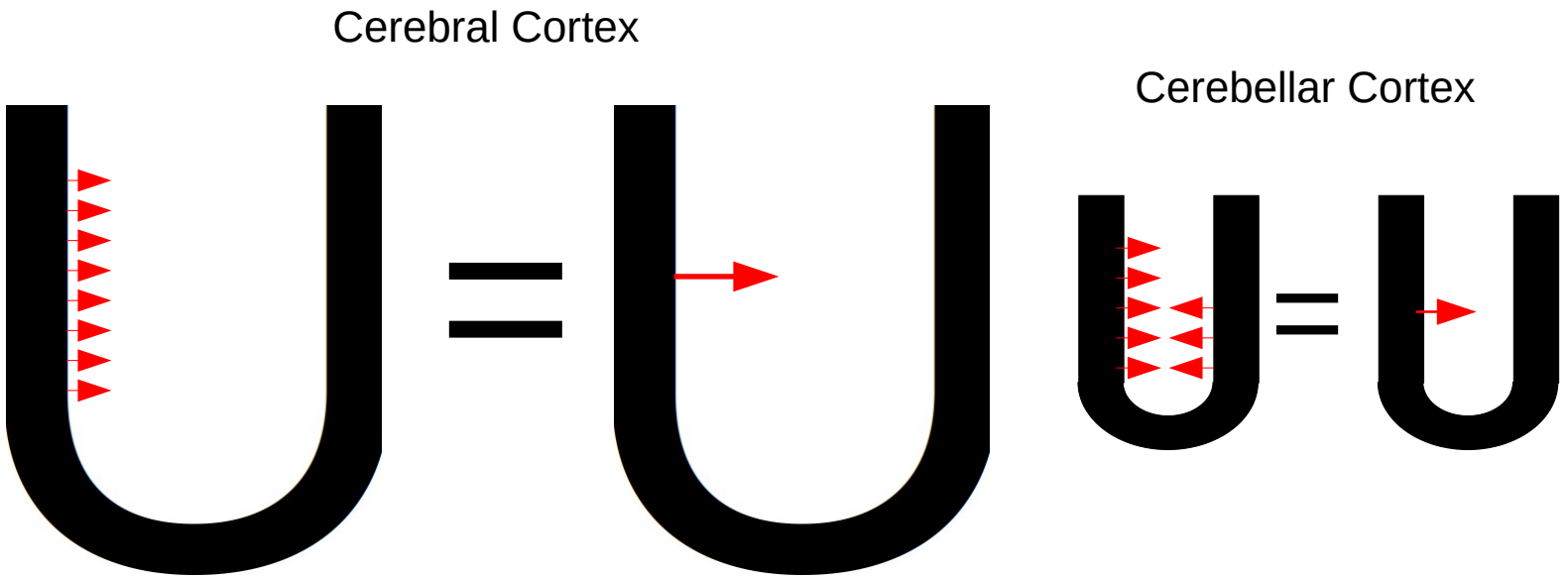
Schematic of summation/cancellation

Sources may cancel one another



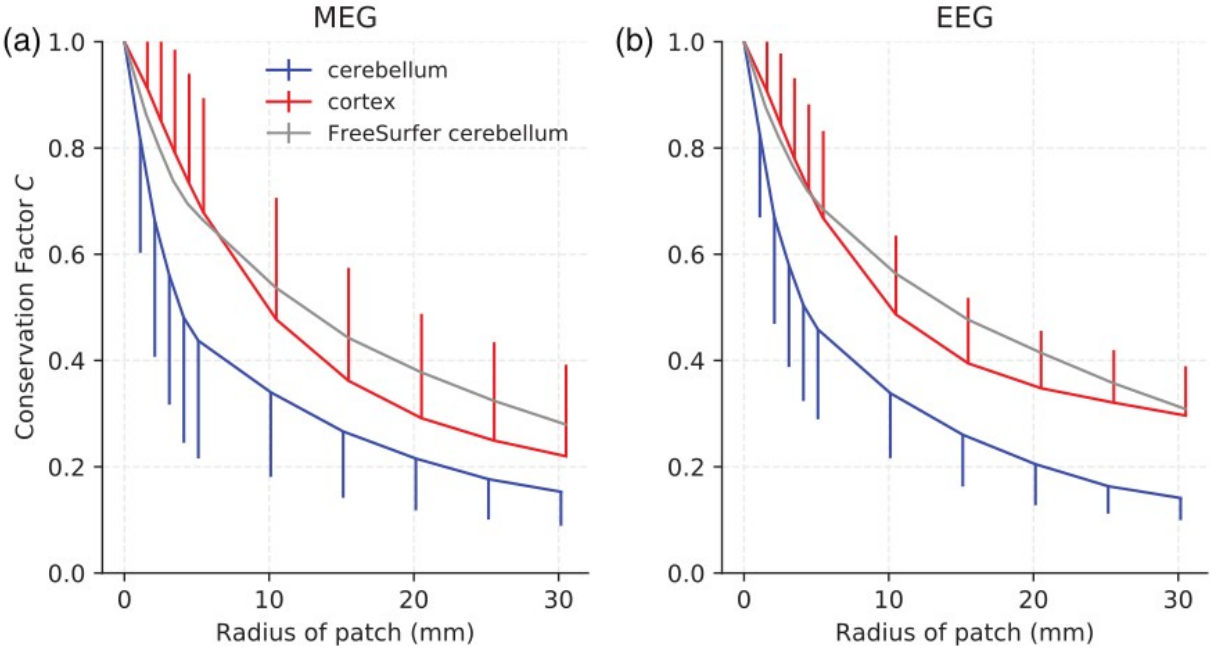
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Schematic of summation/cancellation



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FreeSurfer overestimates the cerebellar signal ...

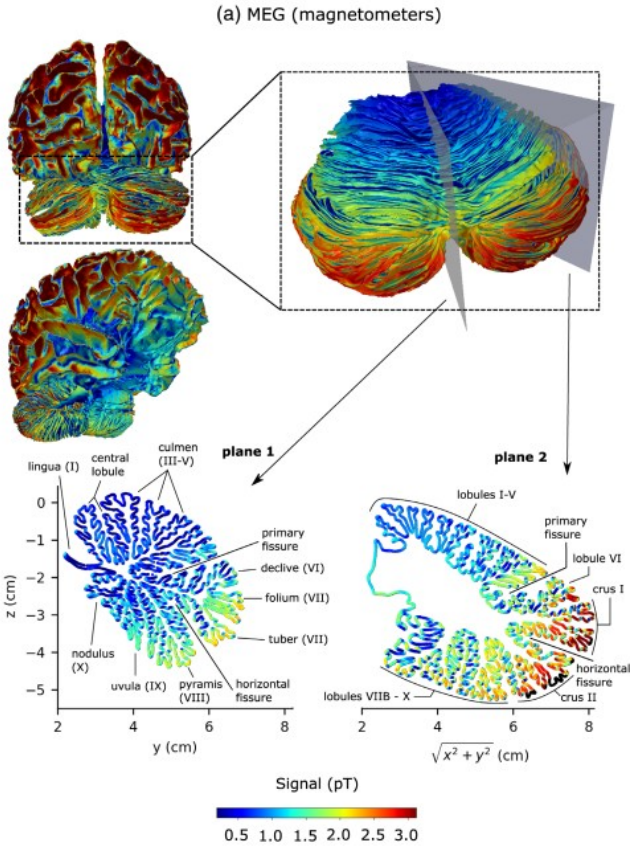


Advice (for now):
use volumetric source models

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Samuelsson et al. 2020

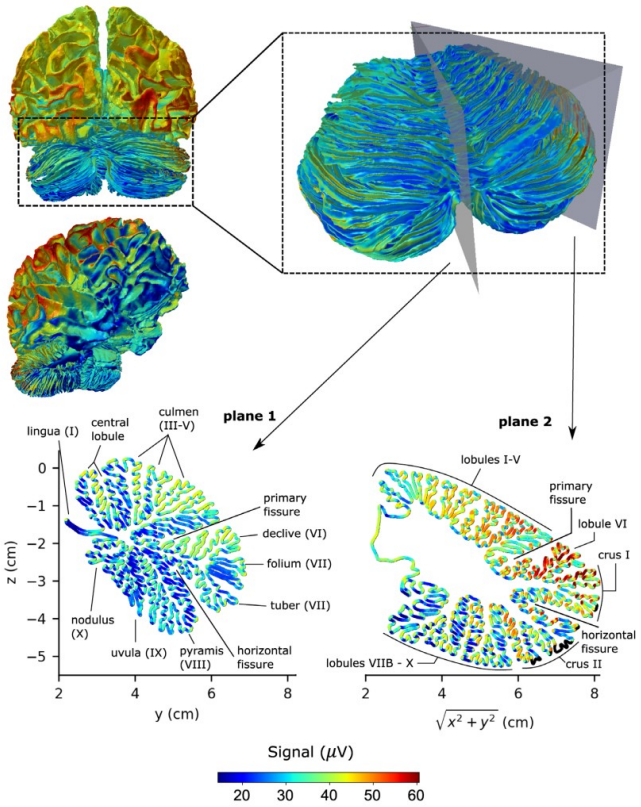
... but cerebellum is within the sensitivity range of MEG



100 nAm dipoles ≈ patch with 10 mm radius and C = 0.3 (conservation factor)

Samuelsson et al. 2020

and EEG



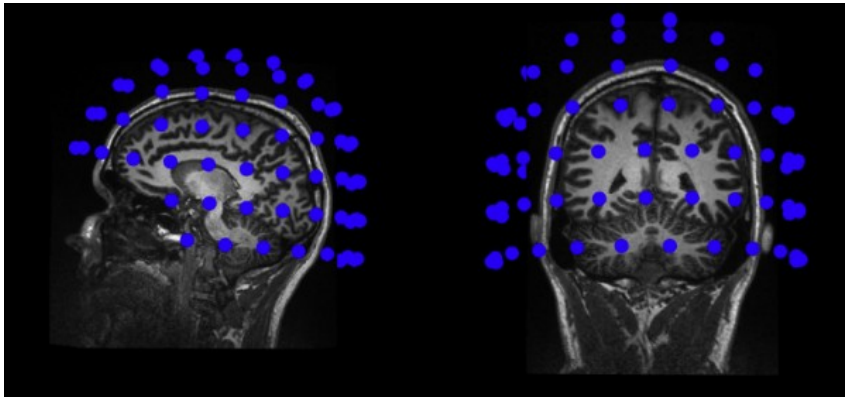
100 nAm dipoles ≈ patch with 10 mm radius and C = 0.3 (conservation factor)

Samuelsson et al. 2020

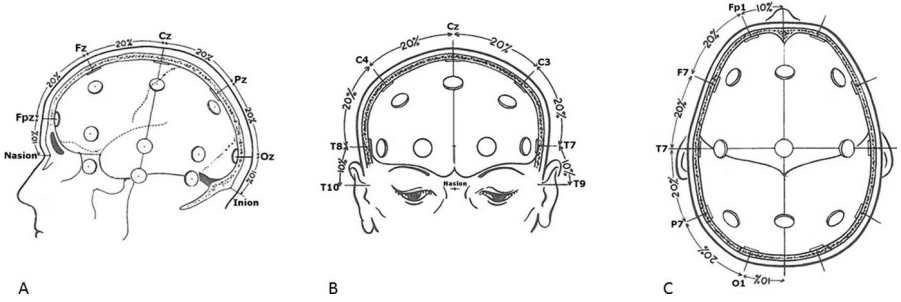
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... but we haven't covered it





Andersen, Jerbi & Dalal 2020



A B C

(Jasper 1958)

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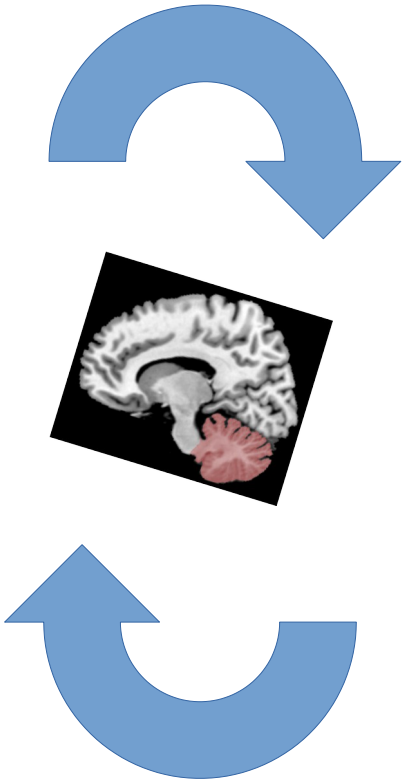
So we basically know very little about the cerebellum, and there aren’t many MEG/EEG studies around.

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TWO IMPEDING MISCONCEPTIONS

1) The cerebellum is primitive
2) The cerebellum cannot be measured non-invasively

A VICIOUS CIRCLE

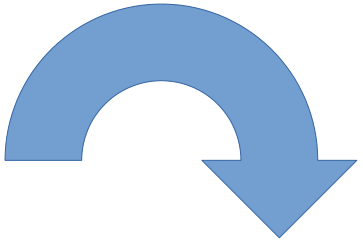


(We think) the cerebellum is primitive; therefore we don’t study it

We do not study the cerebellum; therefore (we think) it is primitive

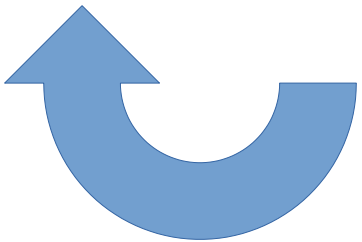
A VICIOUS CIRCLE

—



(We think) the cerebellum is primitive; therefore we don’t study it

Premiss: (We think) we cannot study the human cerebellum; therefore we do not study it and therefore we think it is primitive



We do not study the cerebellum; therefore (we think) it is primitive

INDEPENDENT RESEARCH FUND DENMARK

WHAT WE WILL BE/ARE DOING?



Optically pumped magnetometers
OPM



Respiratory analyses



Deep brain stimulation
(DBS)
and sensing

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RESEARCH GROUP

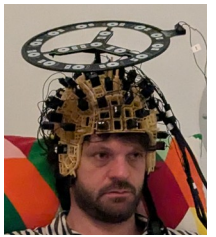
THE (RE)-INTEGRATION OF THE CEREBELLUM

Credit: Laura Paulsen



Lau Andersen
Associate









Laura B.
Paulsen
PhD student



Line Kruse
Postdoc



Anastasia
Weakley
PhD student



MEG: Elekta Neuromag Triux®

WHAT HAVE I ALREADY DONE?

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How I stumbled upon the cerebellum

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ELSEVIER

NeuroImage

Volume 184, 1 January 2019, Pages 78-89



Somatosensory responses to nothing: An MEG study of expectations during omission of tactile stimulations

Lau M. Andersen, Daniel Lundqvist

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https://doi.org/10.1016/j.neuroimage.2018.09.014

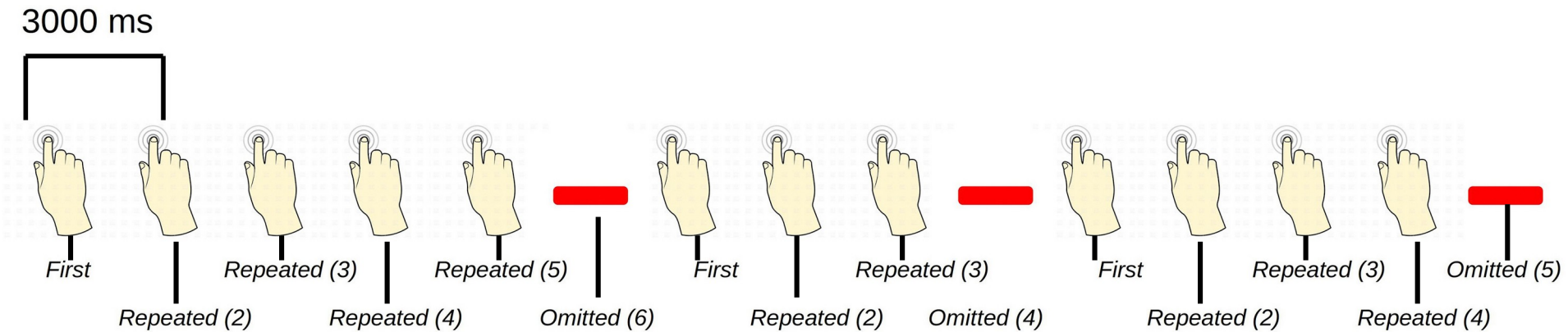
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Paradigm

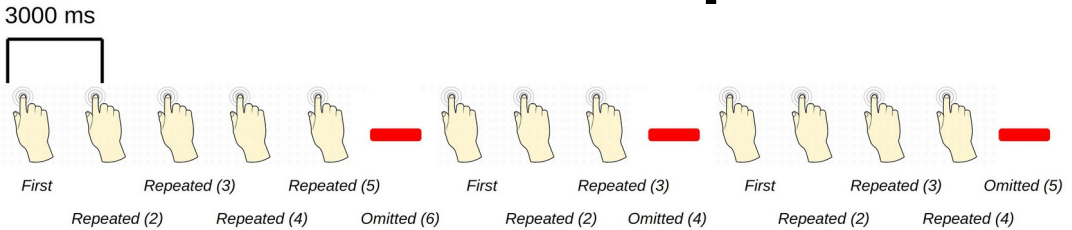


20 subjects
200 first and repeated stimulation
200 omissions

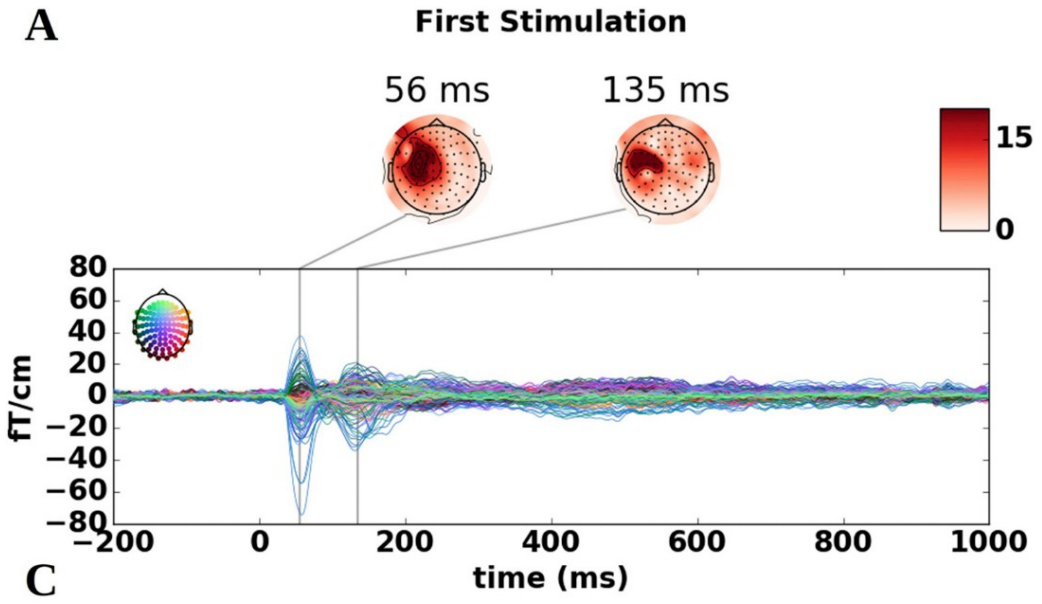
Andersen & Lundqvist 2019

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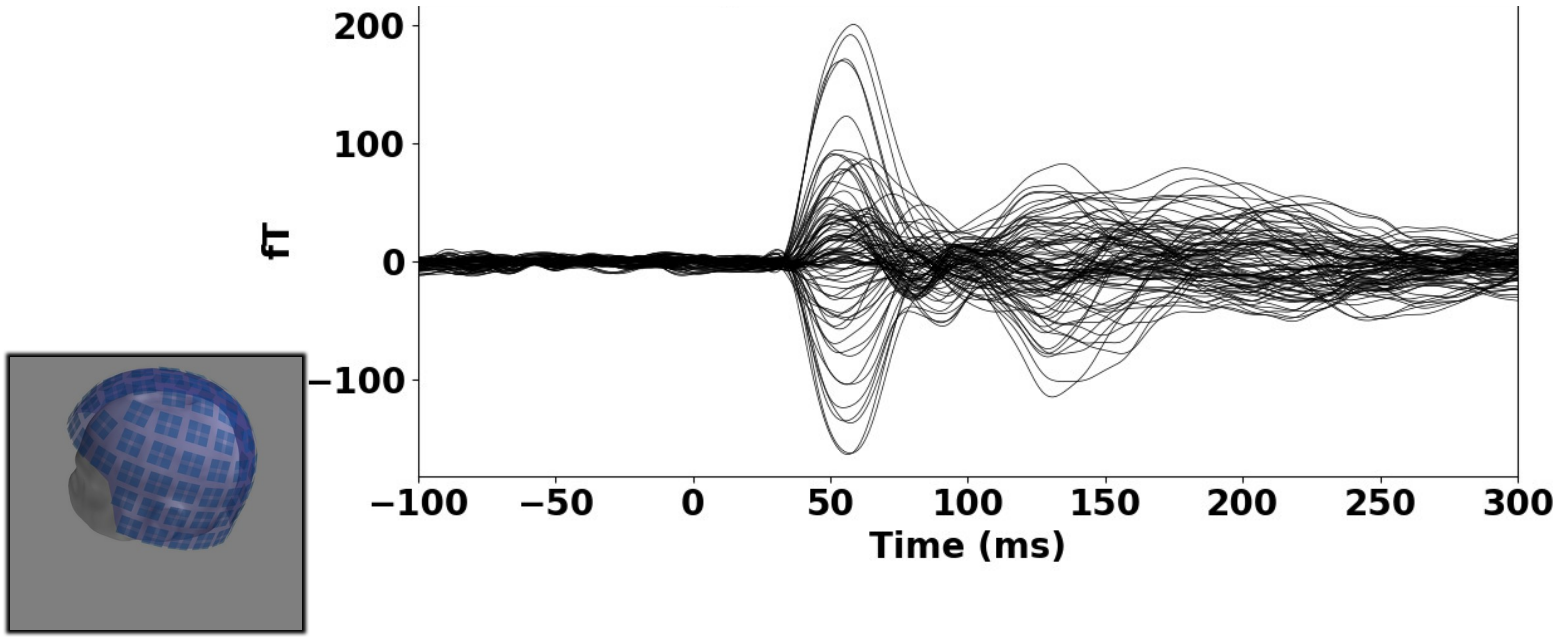
Evoked responses from paradigm



Andersen & Lundqvist 2019

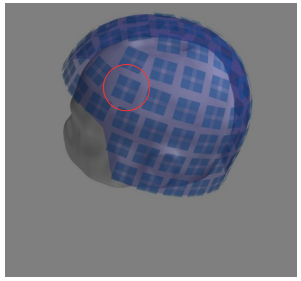


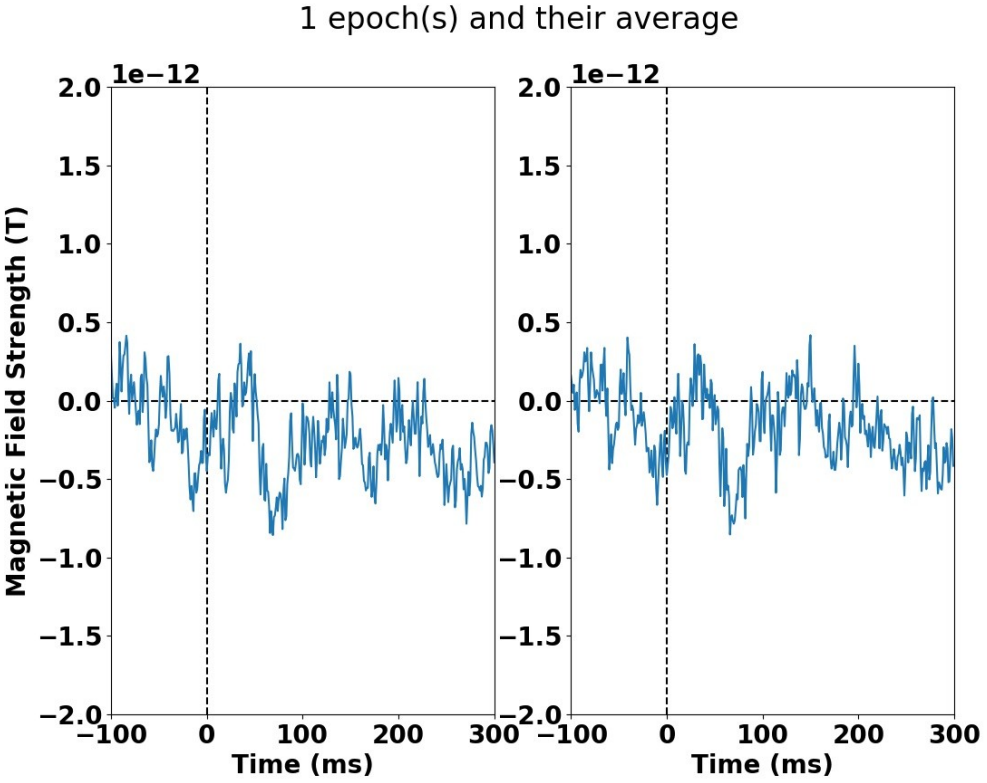
What do signals look like?



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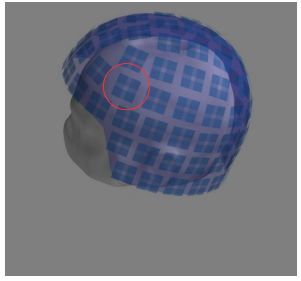
AVERAGES 101

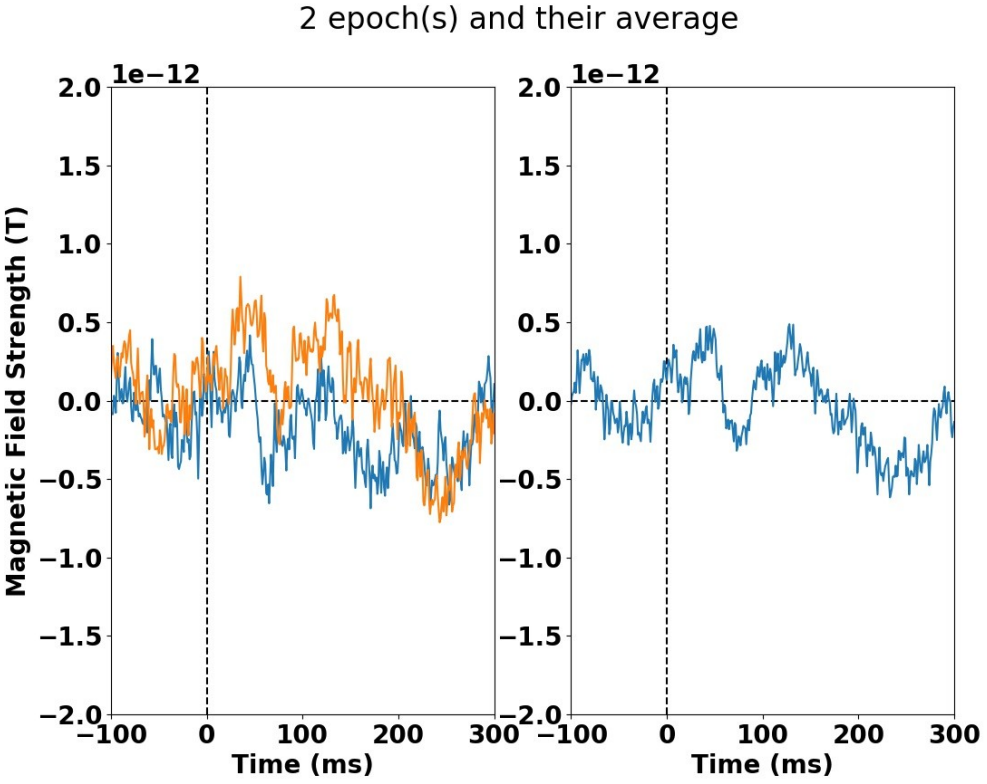




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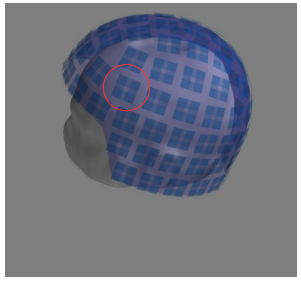
AVERAGES 101

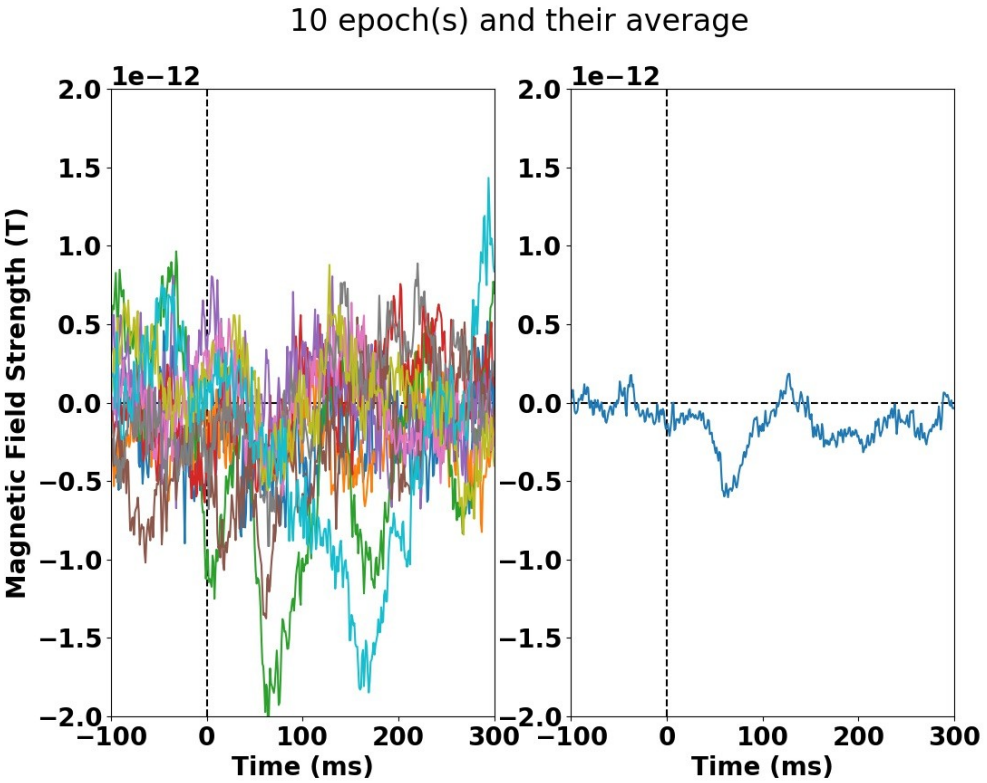




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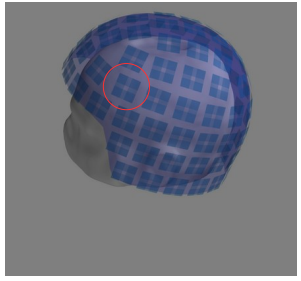
AVERAGES 101

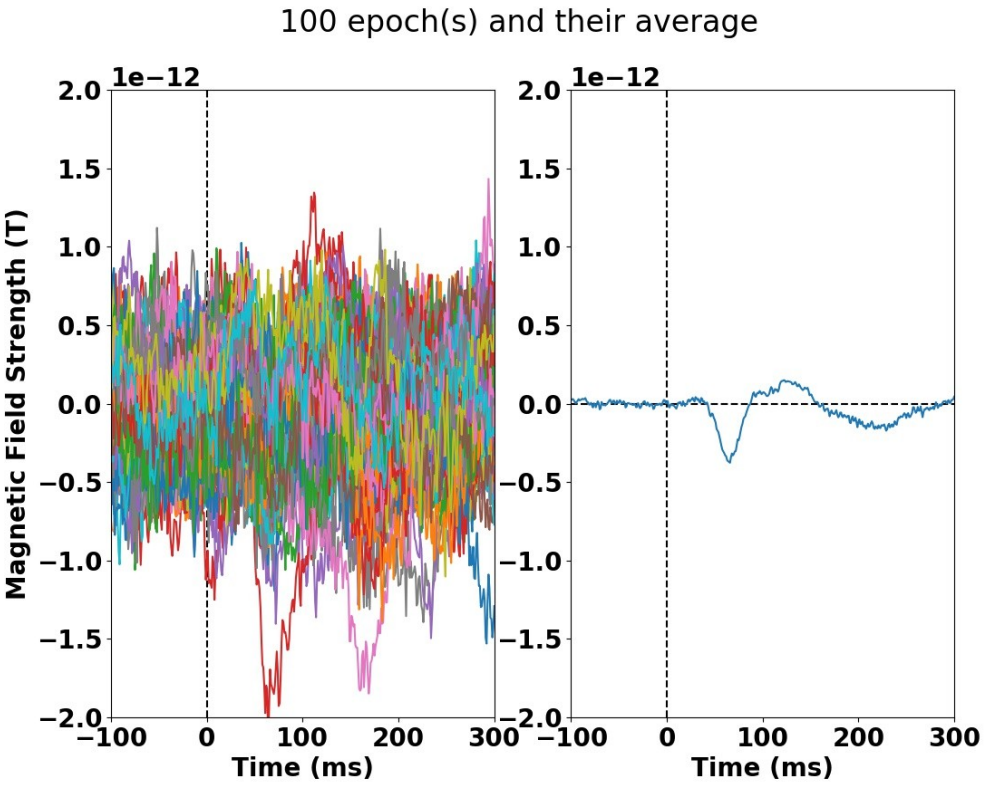




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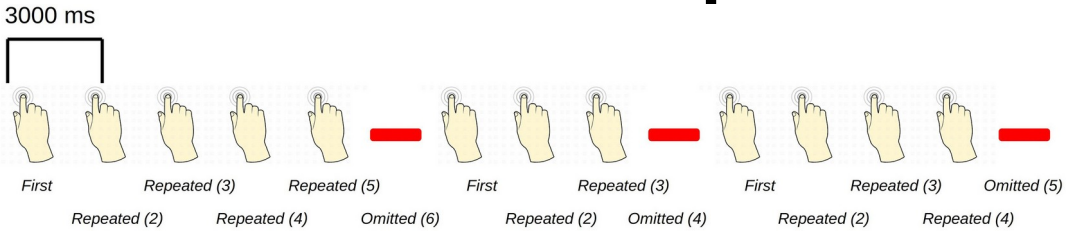
AVERAGES 101

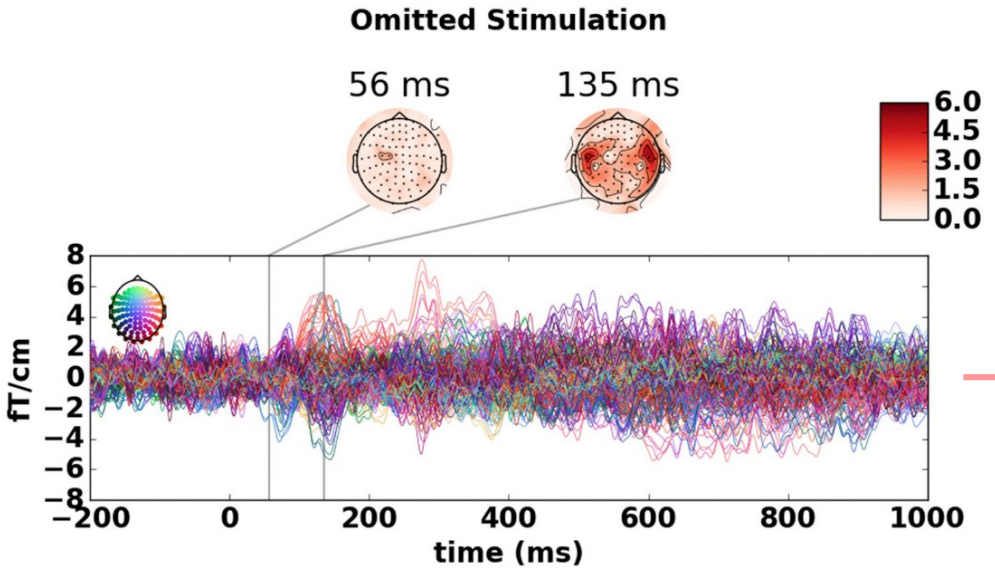




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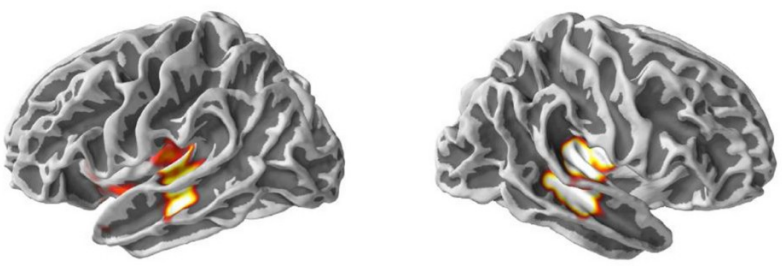
Evoked responses from paradigm





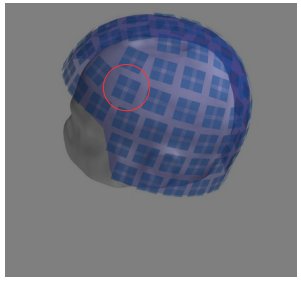
Andersen & Lundqvist 2019

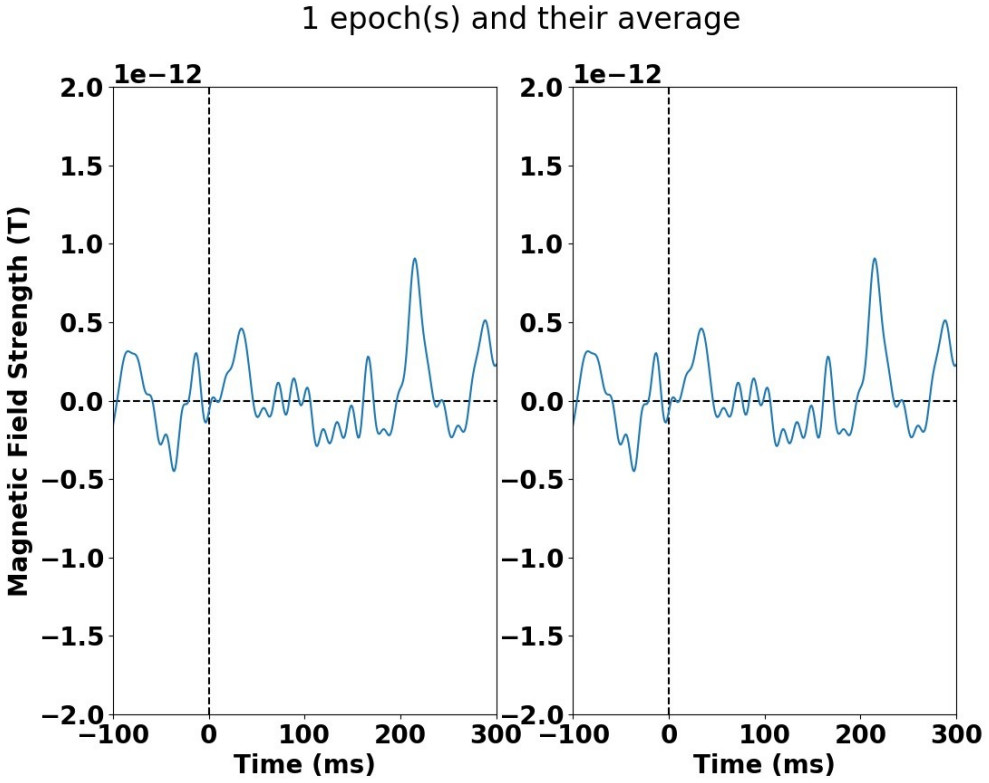
135 ms



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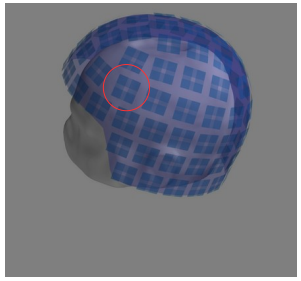
AVERAGES 101- jittered 50 ms

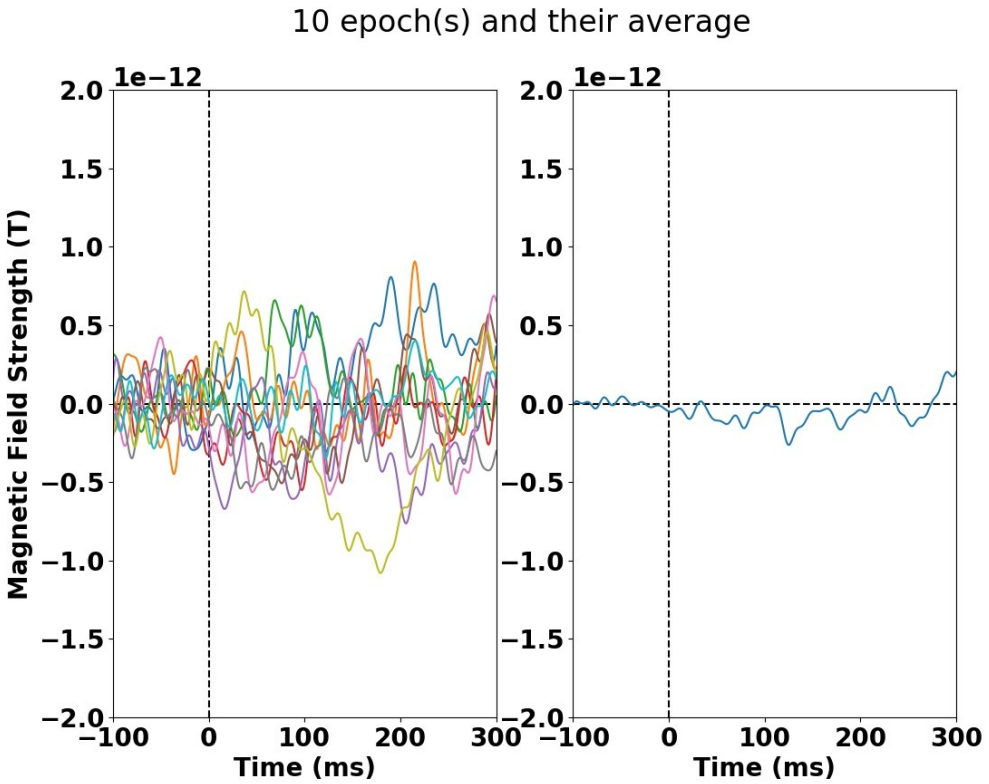




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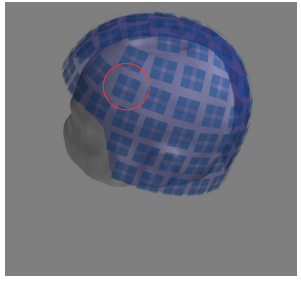
AVERAGES 101- jittered 50 ms

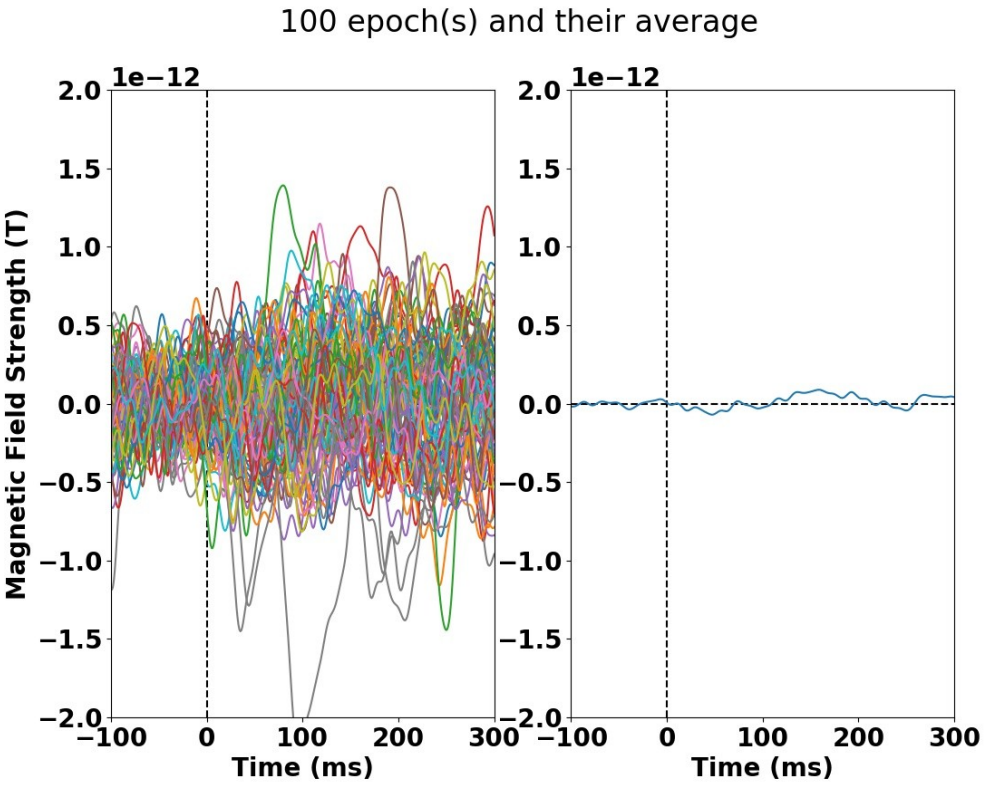




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AVERAGES 101- jittered 50 ms





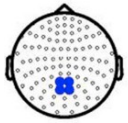
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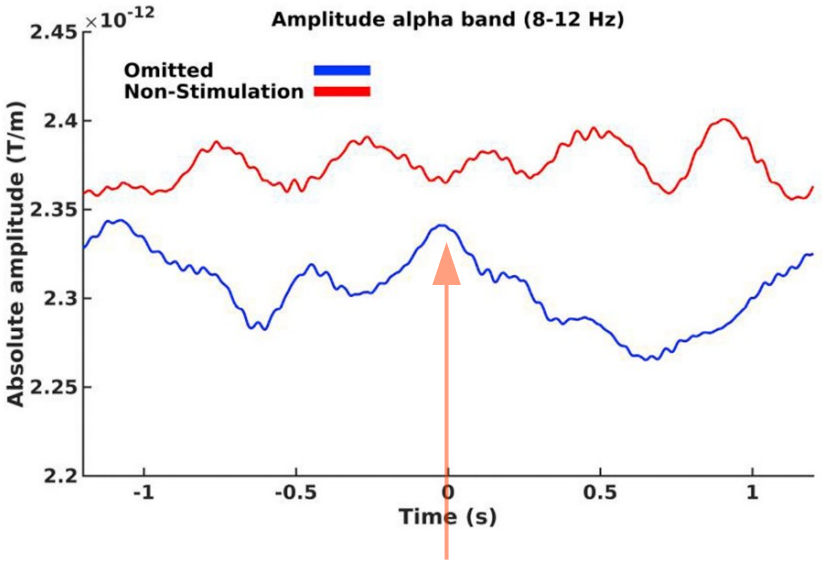
What is time zero for the omissions?

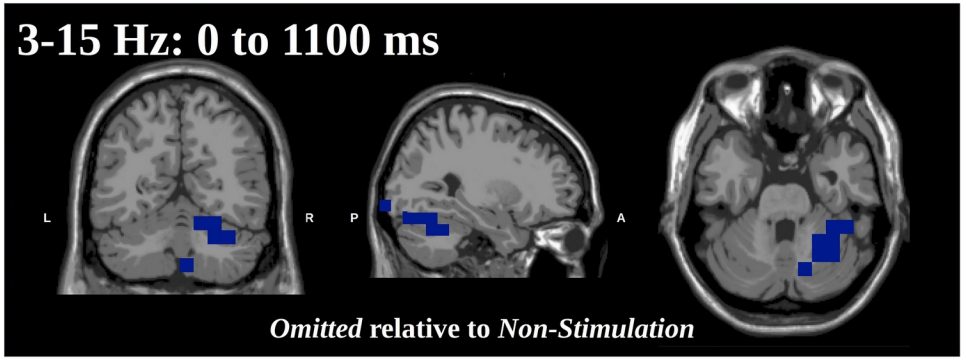
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Non-stimulation differences

Andersen & Lundqvist 2019

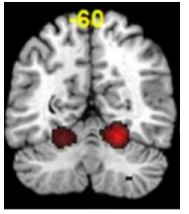






Tonic difference – no time dependency

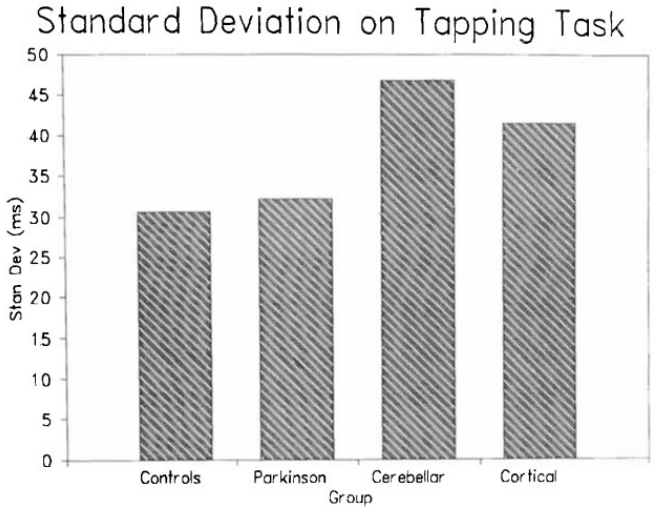
but anything to be made from this peak?!

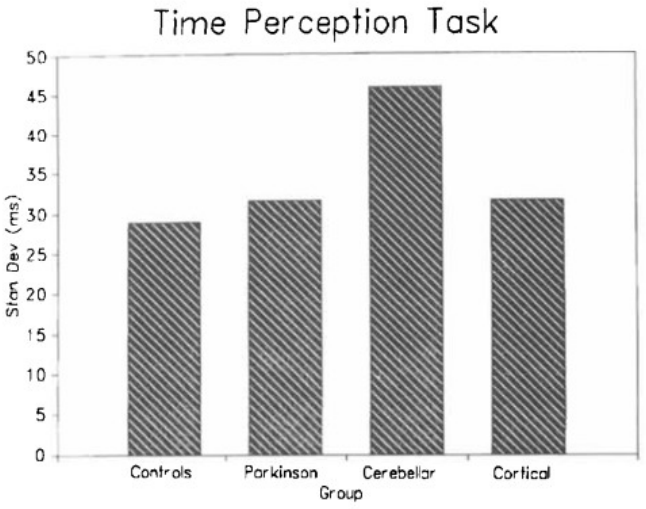


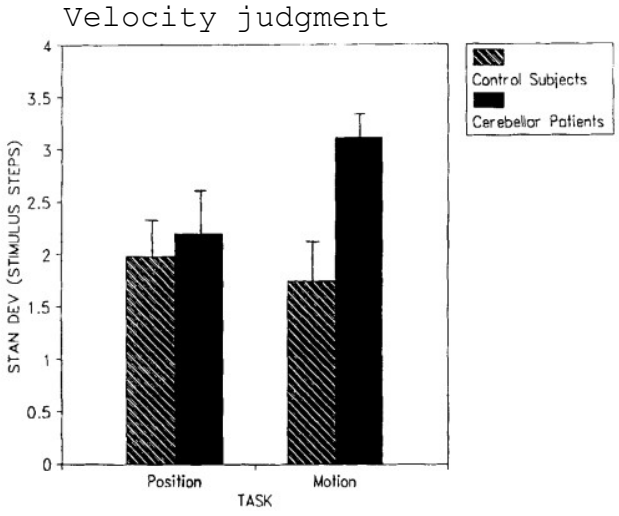
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Keren-Happuch et al. 2014

Cerebellum and timing





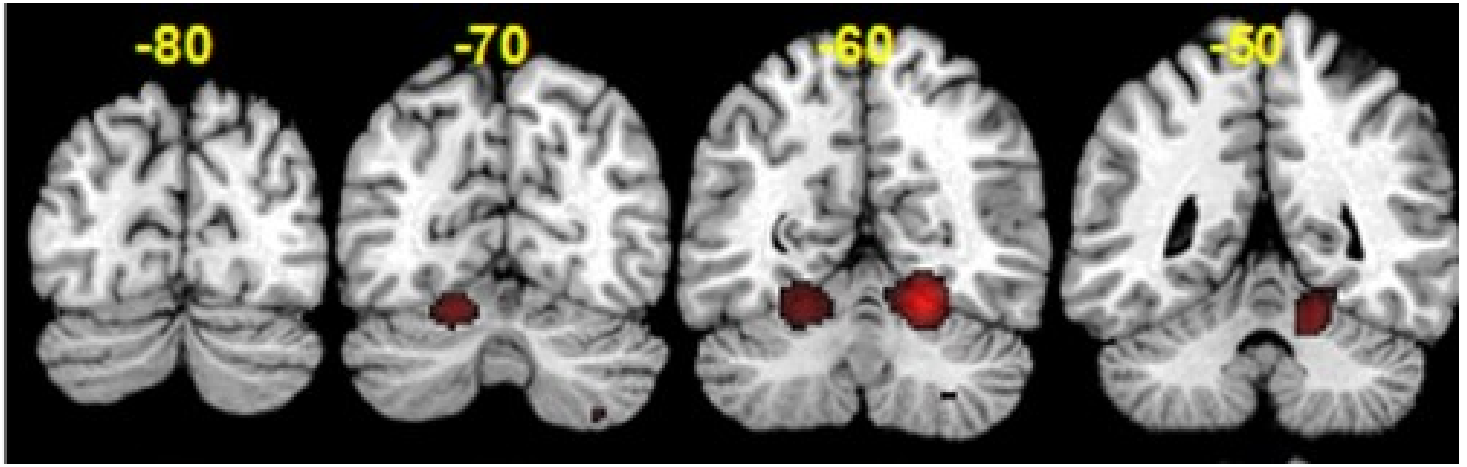


Ivry 1993

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Meta-analysis of 36 fMRI and PET timing studies

Timing



Keren-Happuch et al. 2014

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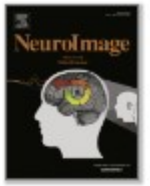
INTERPRETATION

Something is clocking when stimuli should appear?

Is it the cerebellum?

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NeuroImage
Volume 238, September 2021, 118202



The cerebellar clock: Predicting and timing somatosensory touch

Lau M. Andersen [a, b, c], Sarang S. Dalal [a]

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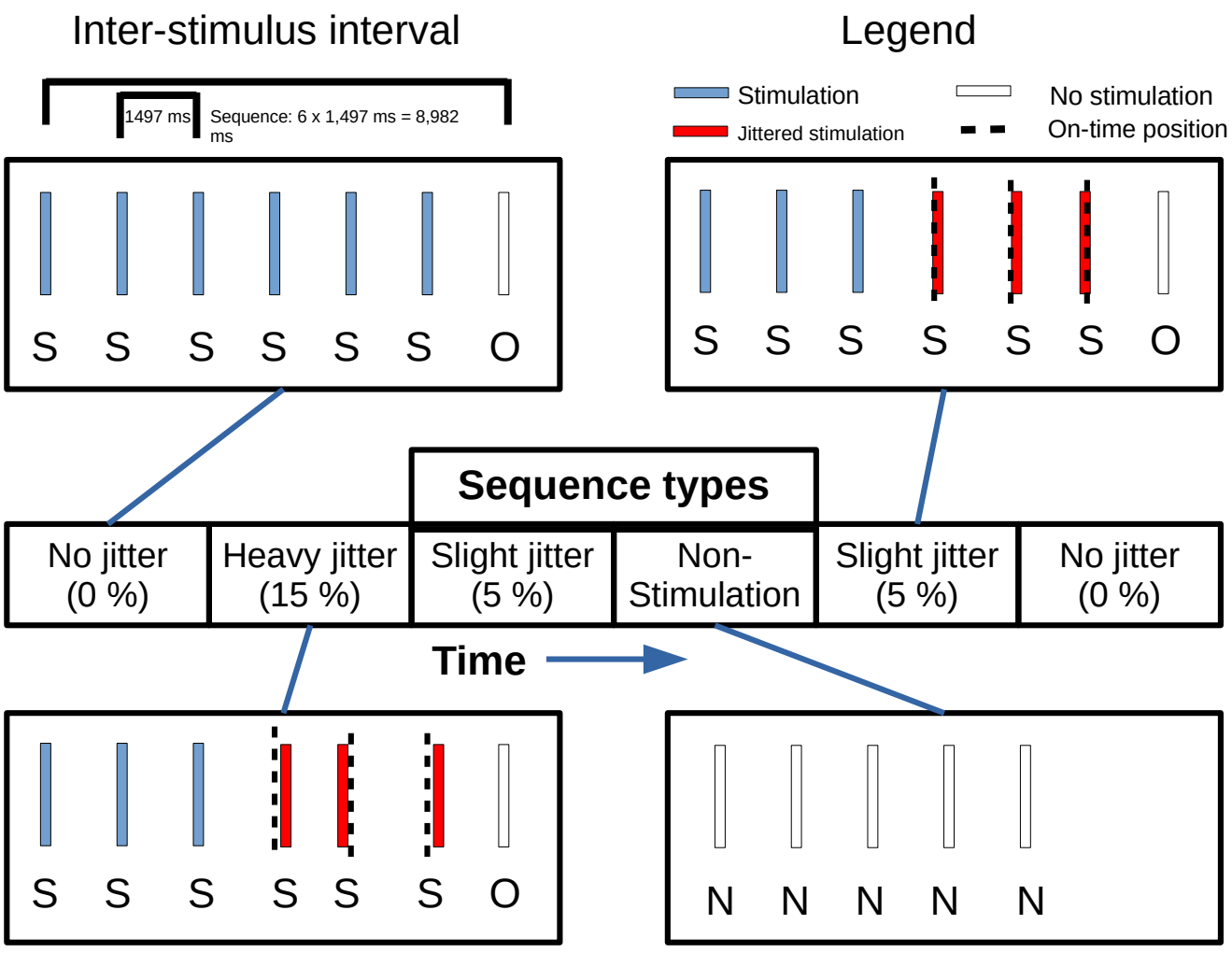
https://doi.org/10.1016/j.neuroimage.2021.118202

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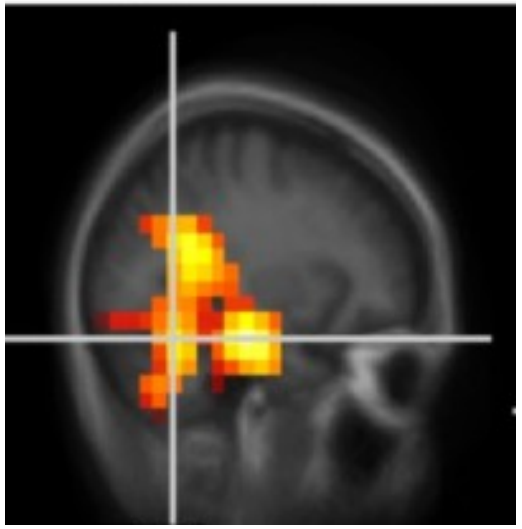
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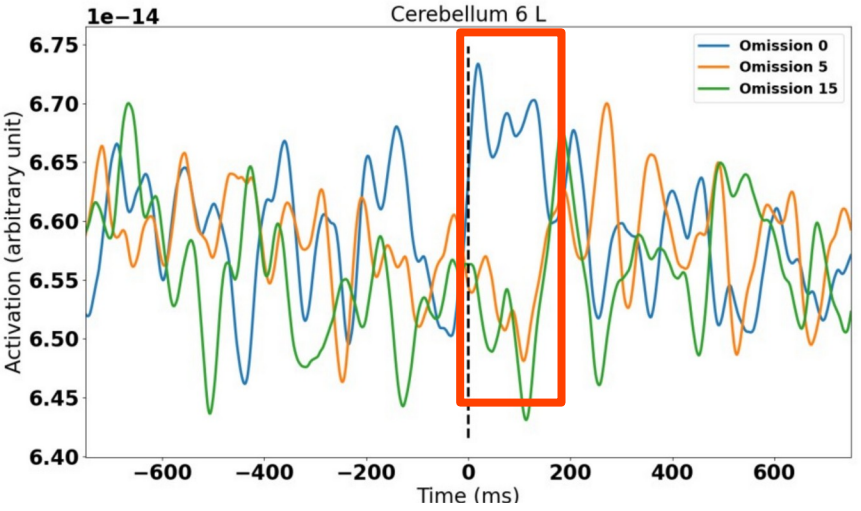
30 subjects
450 first and repeated stimulation
150 omissions of each kind

Andersen & Dalal, *2021*

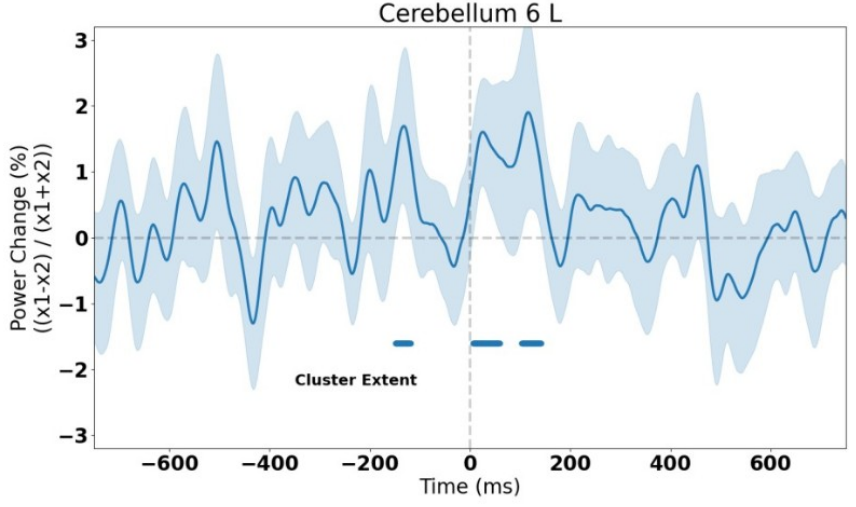
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Regular
Jittered ± 75 ms
Jittered ± 225 ms



Beta band (14-30 Hz)



Andersen and Dalal 2022

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INTERPRETATION

The cerebellum builds expectations that are temporally sensitive

1) How temporally sensitive?
&
2) Are these expectations functionally relevant?

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Kelly Hoogervorst

New Results

The Temporal Constraints of the Cerebellum's Timekeeping

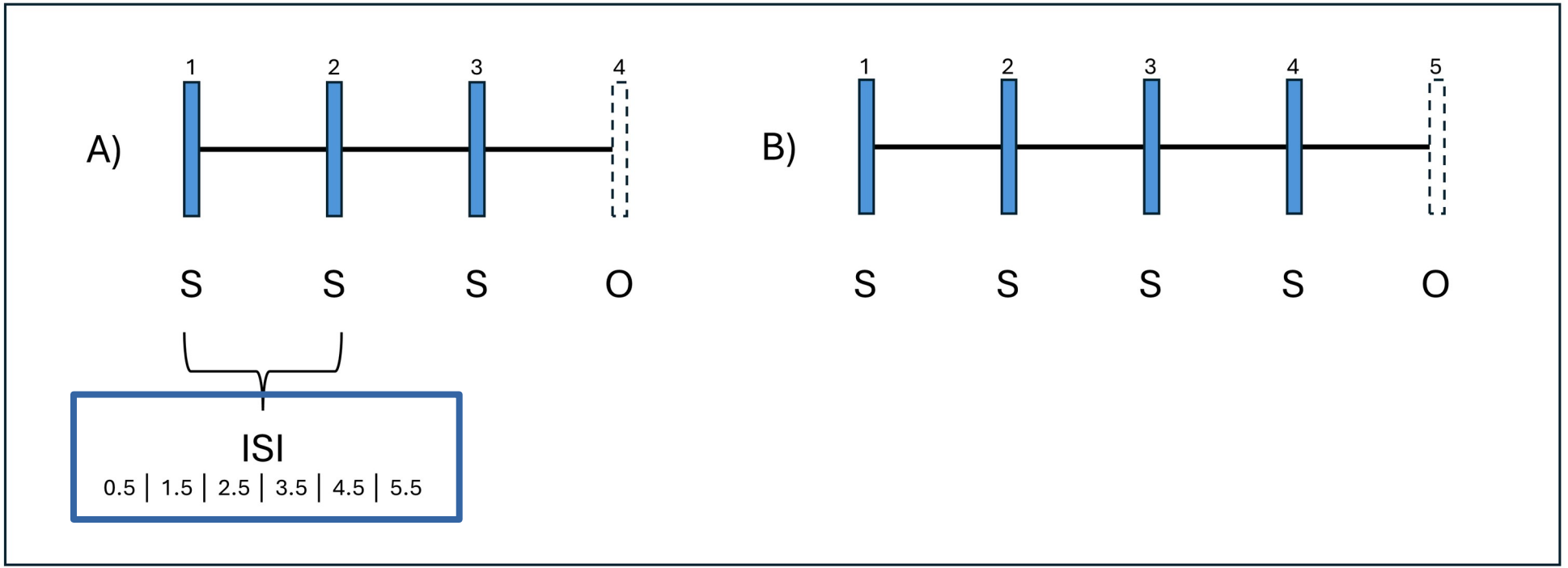
Kelly Hoogervorst, Lau Møller Andersen

doi: https://doi.org/10.64898/2026.04.10.717630

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(Hoogervorst and Andersen, *in revision*)

Paradigm



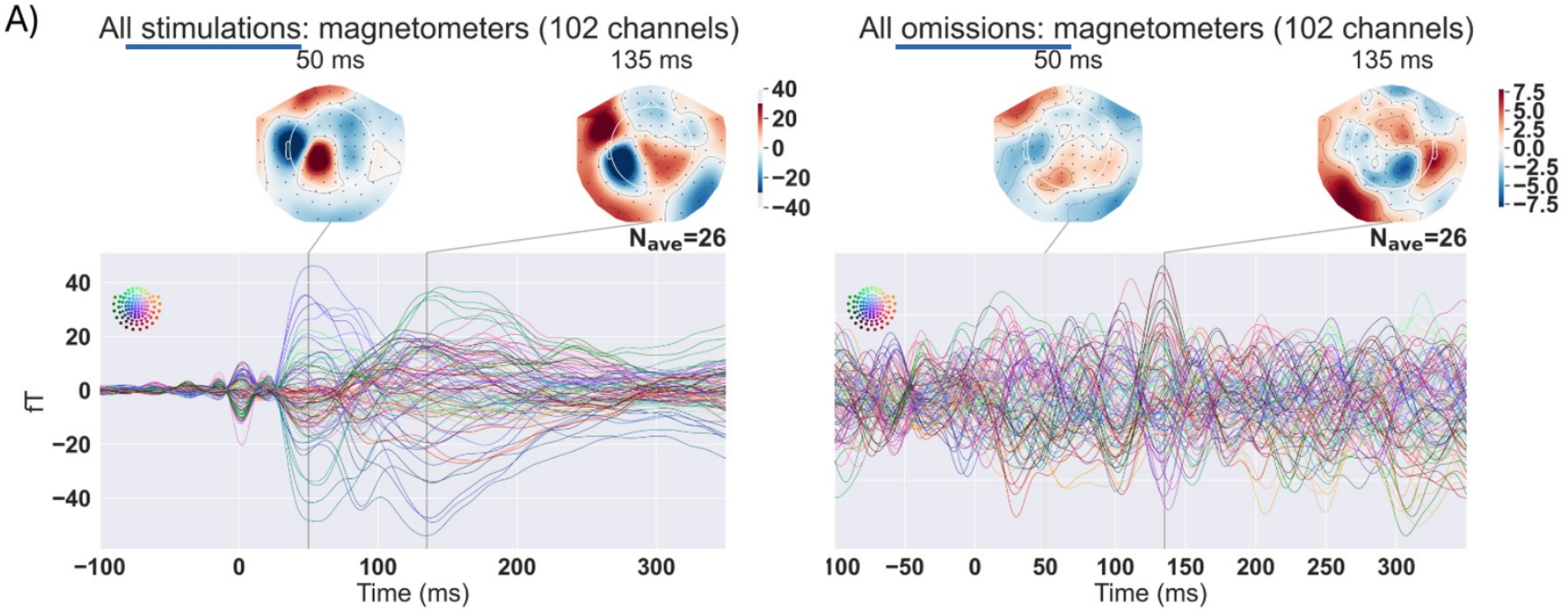
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Hypothesis

Beta band (14-30 Hz) cerebellar responses (~30-50 ms) will diminish in a sigmoid manner as the inter-stimulus interval is increased.

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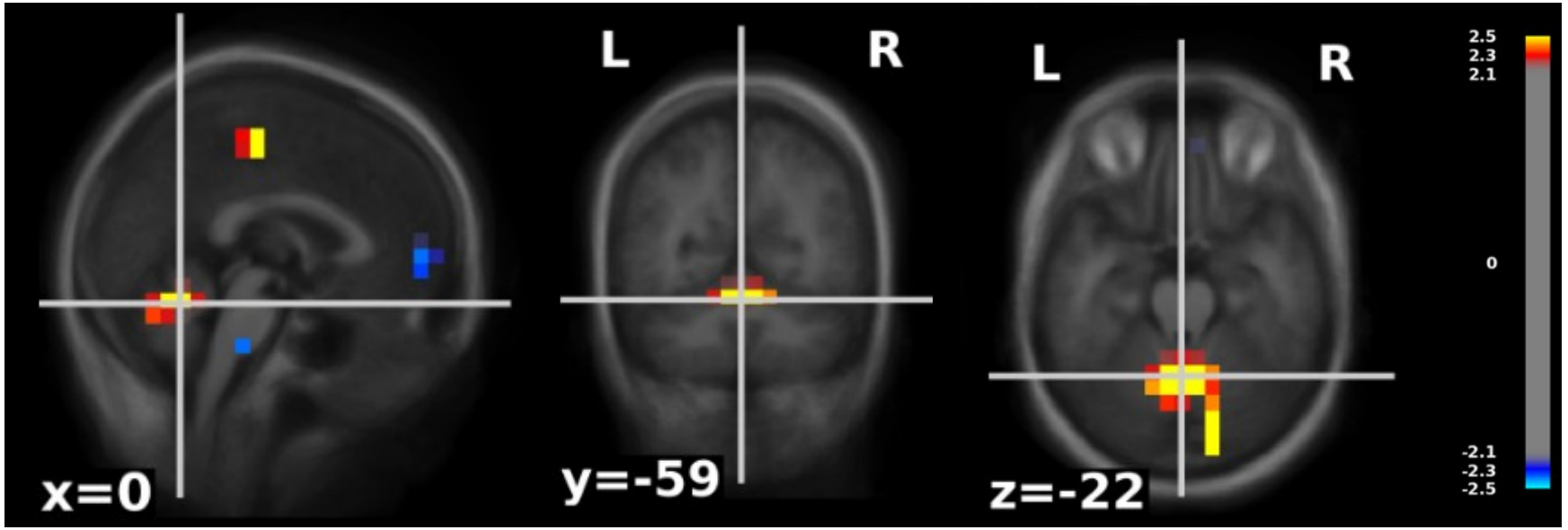
Proof of concept



(Hoogervorst and Andersen, *in revision*)

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Localisation of omissions (collapsed across ISIs; 14-30 Hz)



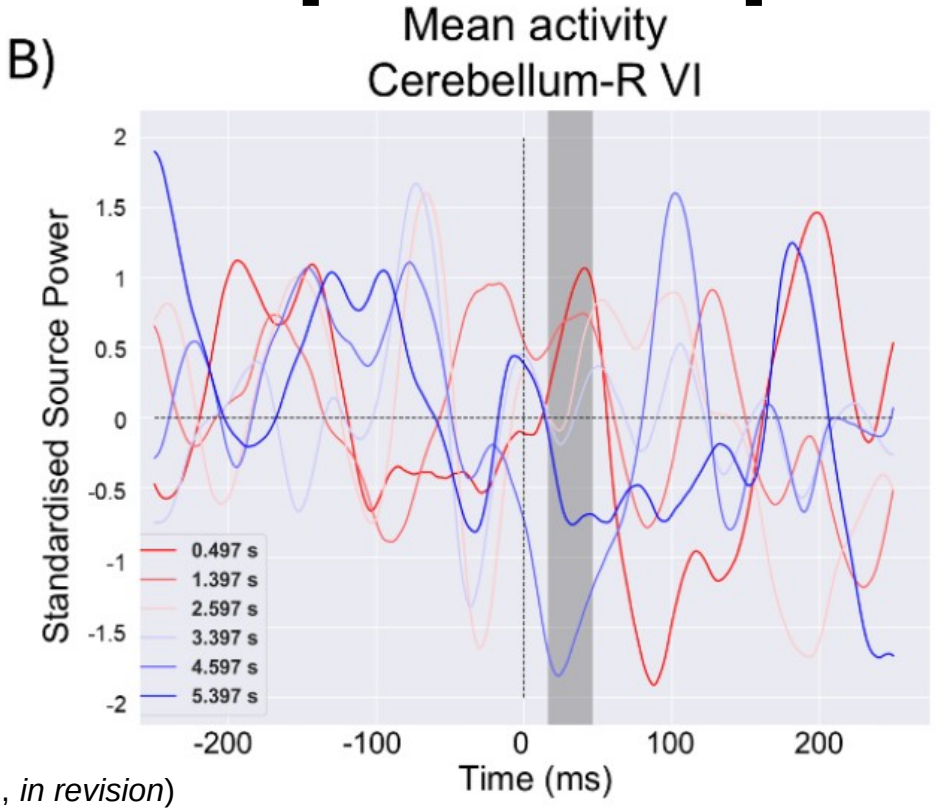
ISI

0.5 | 1.5 | 2.5 | 3.5 | 4.5 | 5.5

(Hoogervorst and Andersen, 2026)

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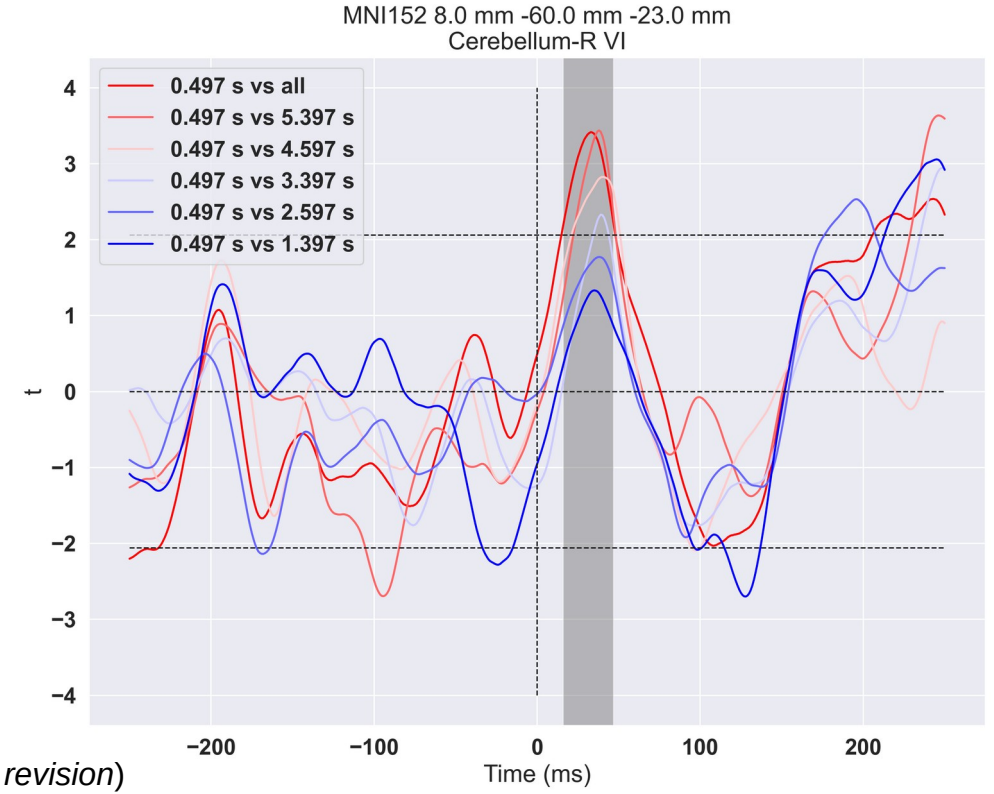
Responses per ISI



(Hoogervorst and Andersen, *in revision*)

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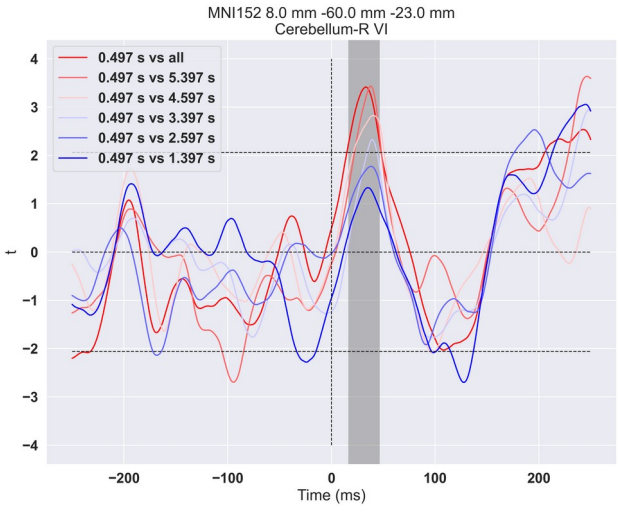
Test against shortest ISI (0.497 s)

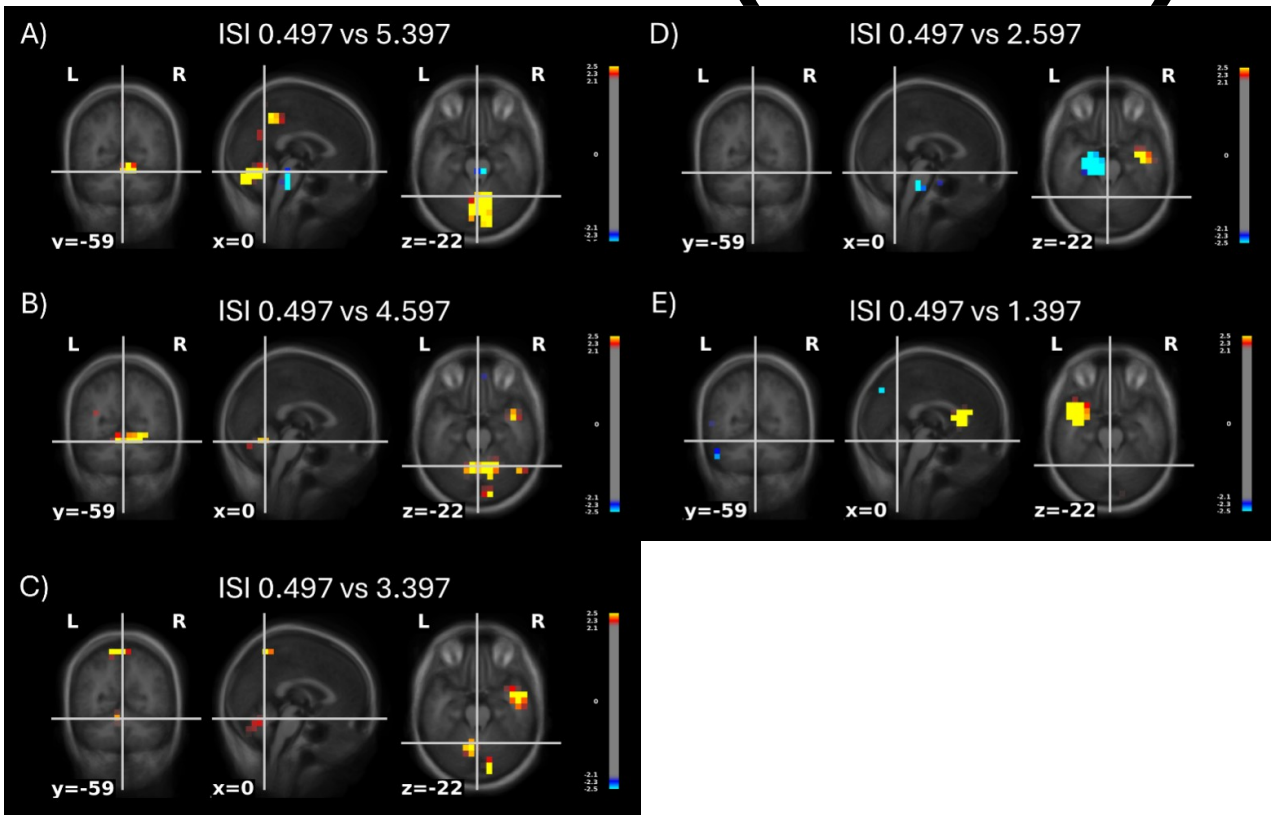


(Hoogervorst and Andersen, *in revision*)

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Test against shortest ISI (0.497 s)





(Hoogervorst and Andersen, *in revision*)

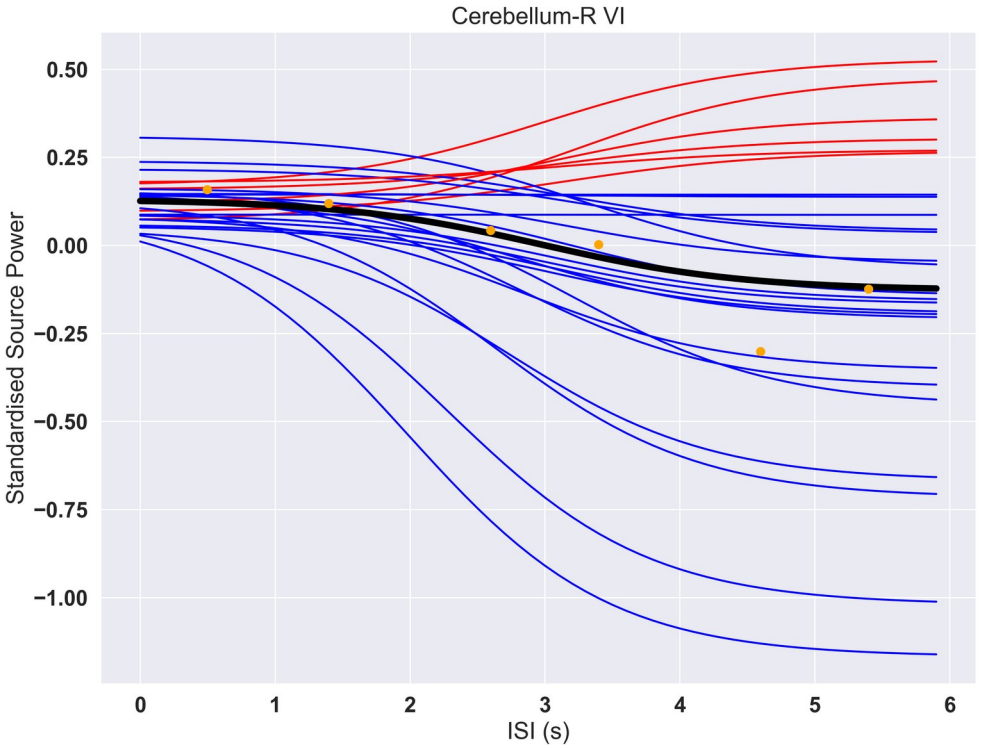
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Logistic fit

$$f(x) = a - \frac{b-a}{1+e^{\frac{c-x}{d}}}$$

a: lower asymptote
b: upper asymptote
c: inflexion point
d: steepness
e: Euler's number
x: ISI (s)
f(*x*): Modelled cerebellar source power

(Hoogervorst and Andersen, *in revision*)



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QUESTION:
What interval durations can the cerebellum keep track of?

(STRONG) ANSWER:
Durations < 4 s

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Future questions

- ***Functional relevance***
  - **Predictions:**
    - Up to durations of 4 s, behavioural performance should depend **linearly** on ISI
    - After 4 s, behavioural performance relative to ISI, should undergo a **qualitative** shift

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Now back to
Are these expectations functionally relevant?

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eNeuro

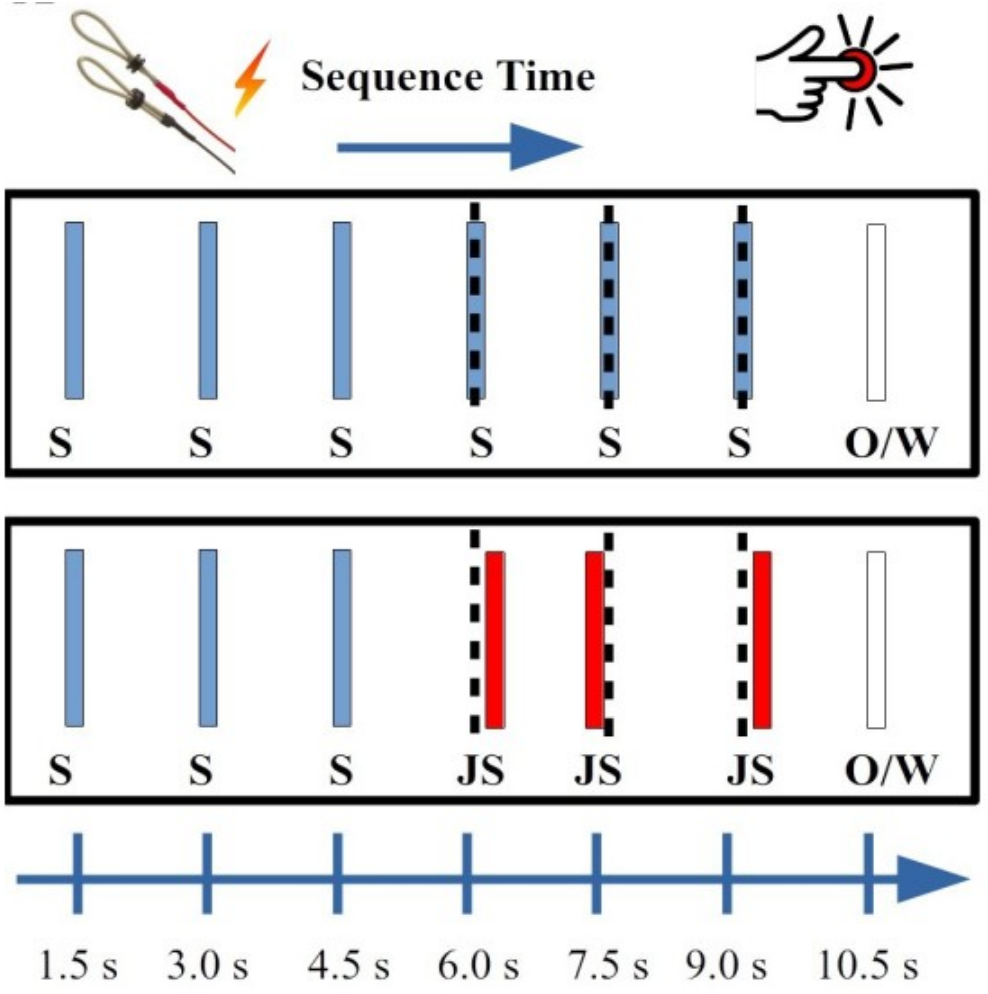
Research Article: New Research
Cognition and Behavior

Detection of Threshold-Level Stimuli Modulated by Temporal Predictions of the Cerebellum

Lau M. Andersen[1,2,3] and Sarang S. Dalal[1]

[1]Center of Functionally Integrative Neuroscience (CFIN), Aarhus University, Aarhus C 8000, Denmark, [2]Aarhus Institute of Advanced Studies (AIAS), Aarhus University, Aarhus C 8000, Denmark, and [3]Department for Linguistics, Cognitive Science and Semiotics, Aarhus University, Aarhus C 8000, Denmark

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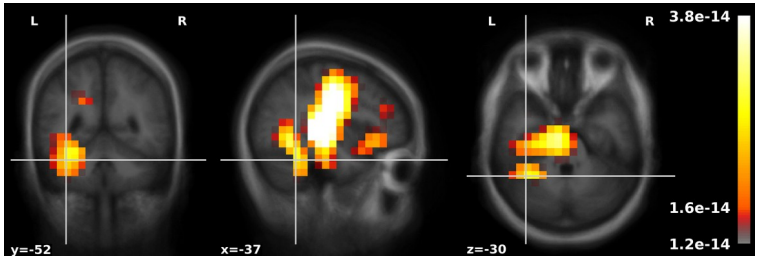


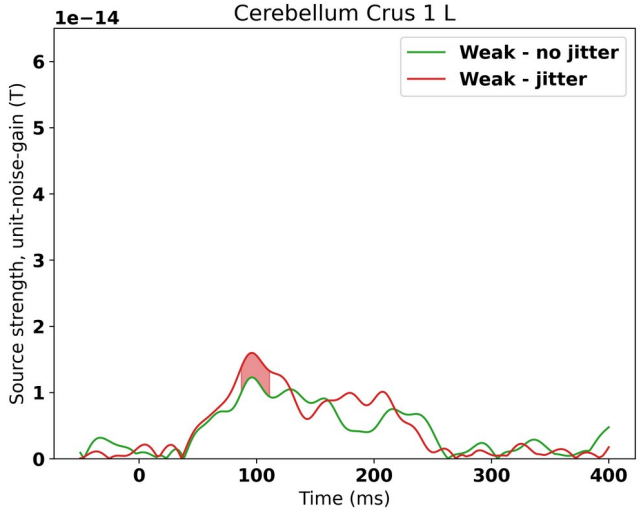
30 subjects

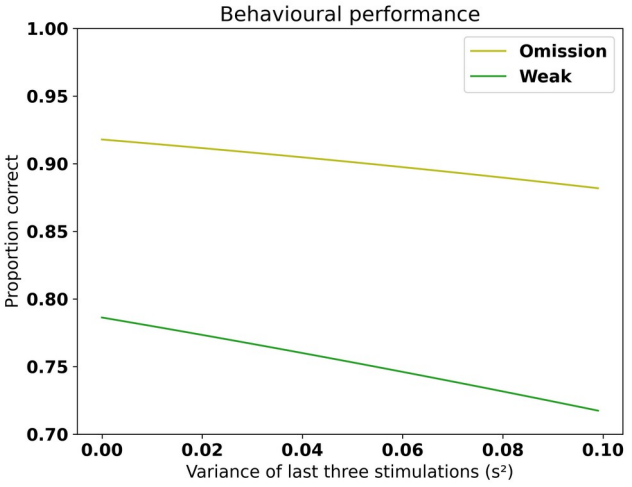
n trials:

- 75 Weak Jittered
- 75 Weak regular
- 75 Omission jittered
- 75 Omission regular

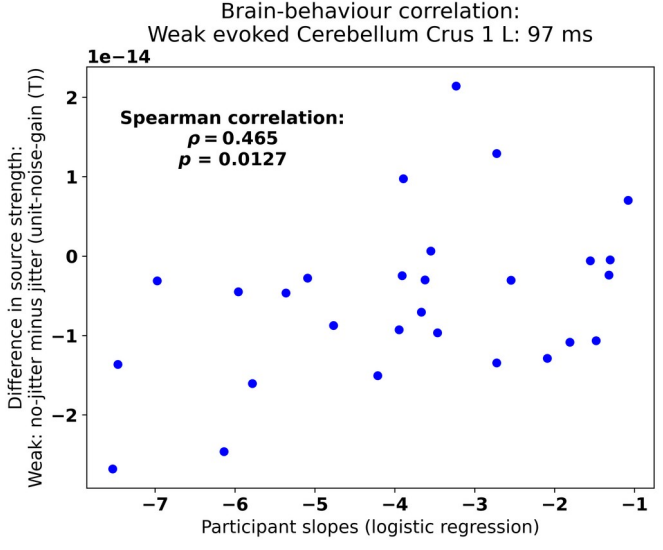
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$$\sigma^2_{\text{STIMULATION}} = \sum_{i=4}^{n=6} \left(t^{(i)}_{\text{ACTUAL}} - t^{(i)}_{\text{EXPECTED}}\right)^2,$$
$$\text{with: } t^{(i)}_{\text{EXPECTED}} = 1,497\,\text{ms} \times i$$

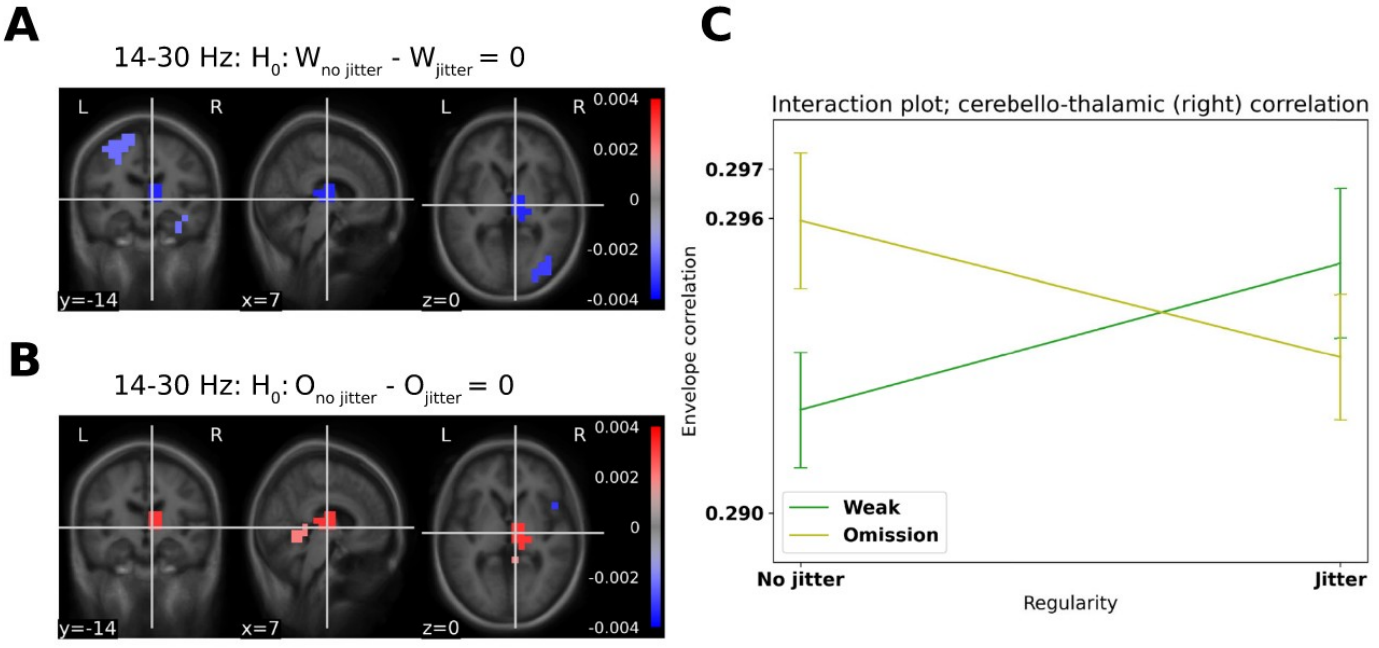


Andersen & Dalal, 2024

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Beta band connectivity

-100 ms to 100 ms



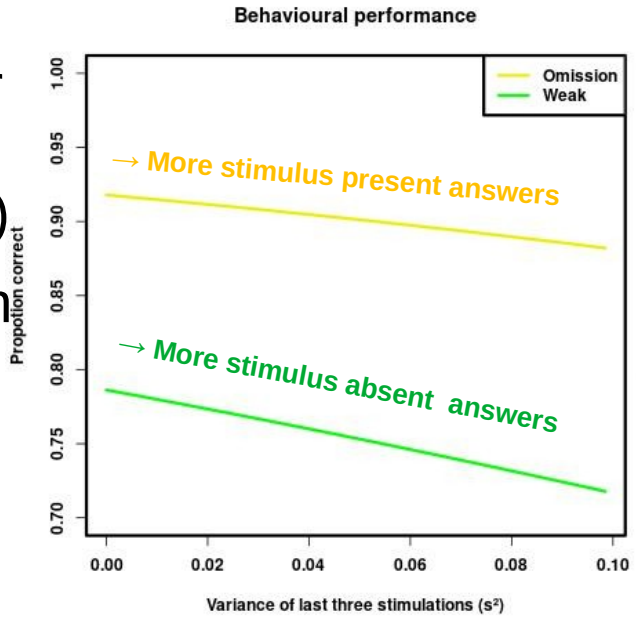
$F_{1,100} = 4.42$, $p = 0.0376$

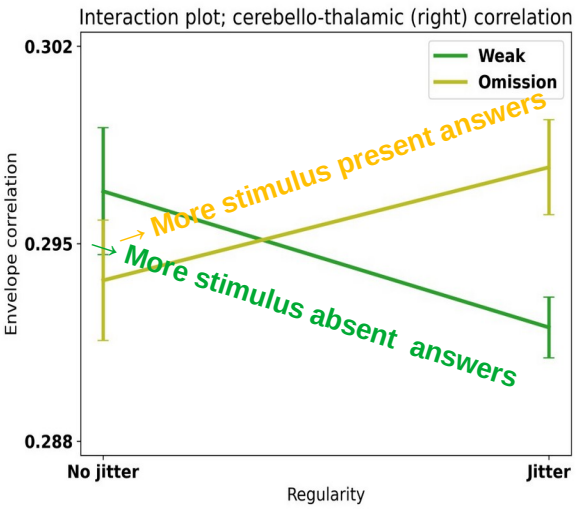
Andersen & Dalal, 2024

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RIGHT THALAMUS

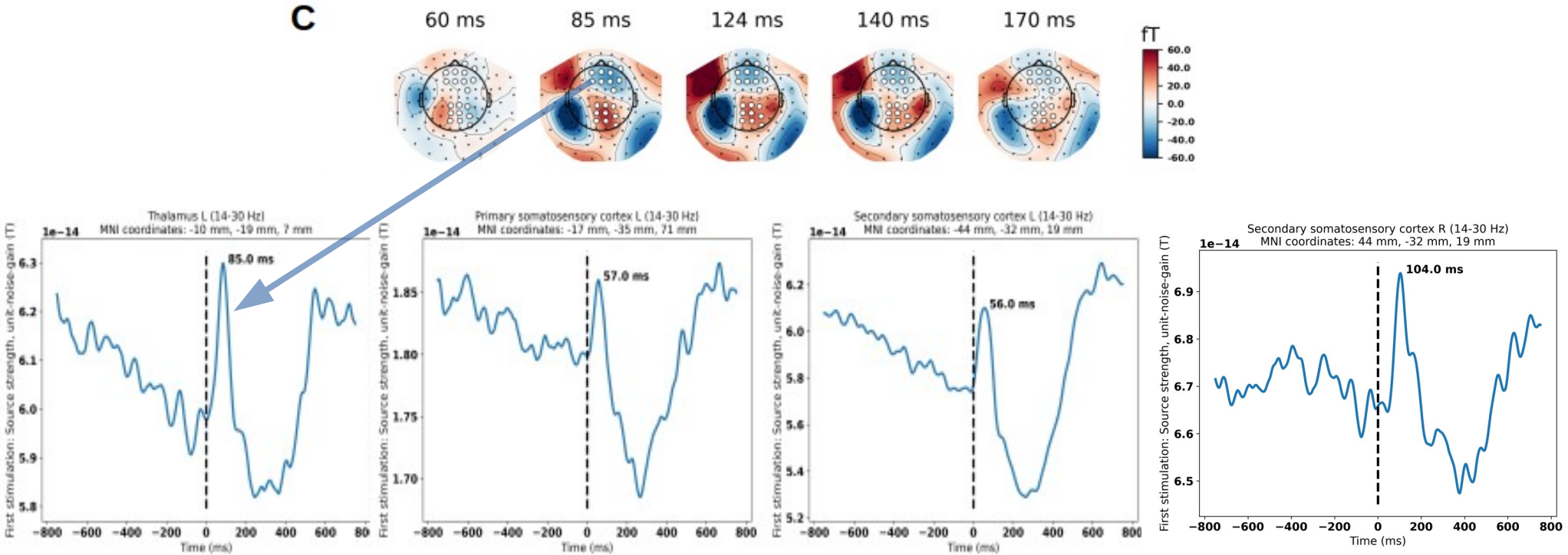
- Subjects were to deliver their answers with their **left** hand (connected to right thalamus)
- This fits with an interpretation of cerebello-thalamic correlation increasing when subjects answer “stimulus present”





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IS IT REALLY THE THALAMUS?



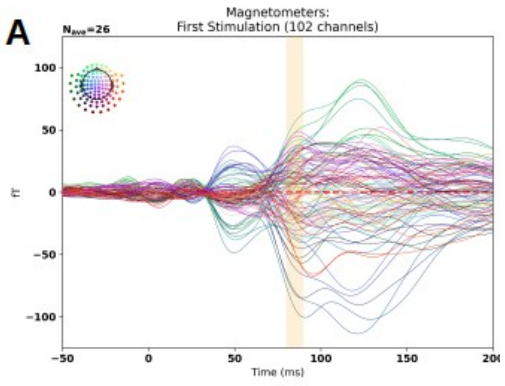
Andersen & Dalal, 2024

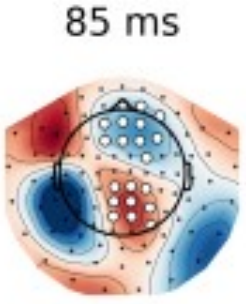
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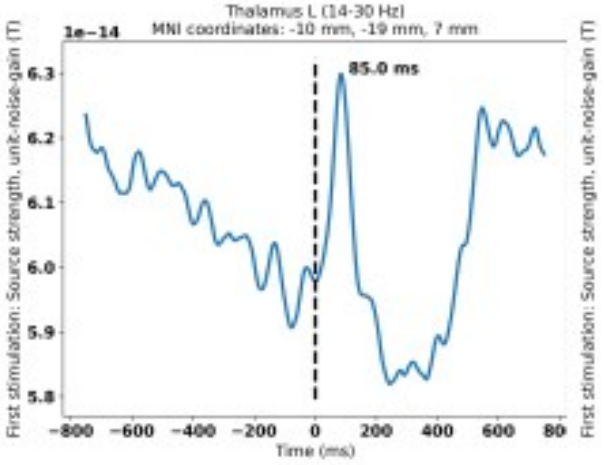
MAIN ARGUMENTS FOR DEEP ACTIVITY

1) the thalamic peak response at 85 ms for the beta band (14-30 Hz) coincides with an evoked response at 85 ms visible in sensor space

2) the topographical pattern at 85 ms for the magnetometers is compatible with a central, deep source







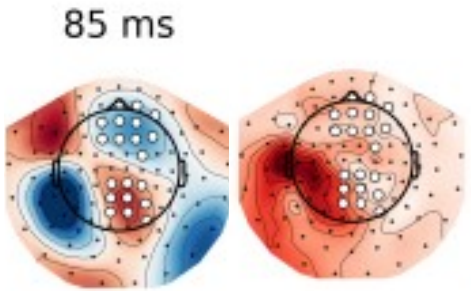
Andersen & Dalal, 2024

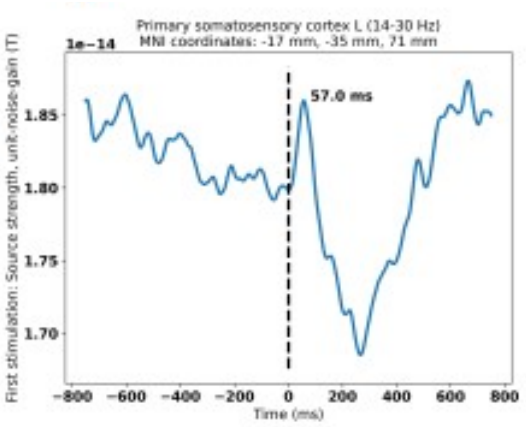
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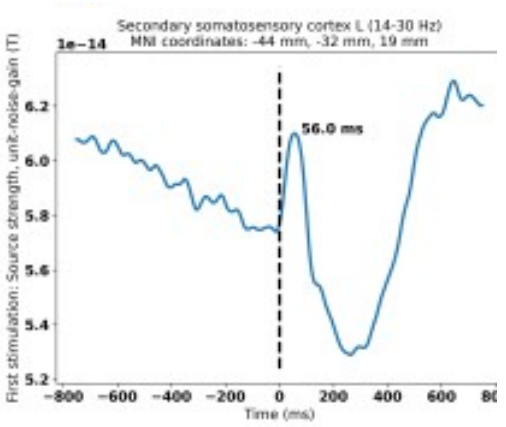
MAIN ARGUMENTS FOR DEEP ACTIVITY

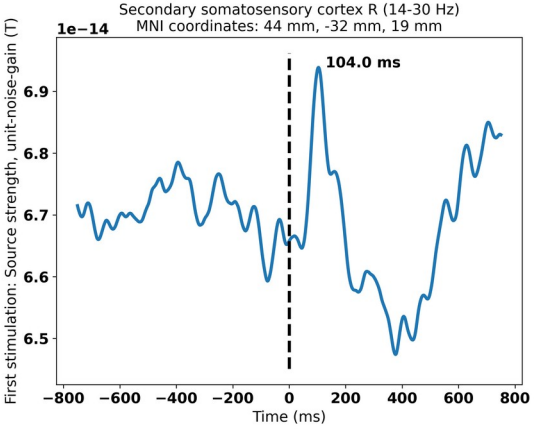
3) the topographical pattern at 85 ms (or later) for the planar gradiometers does not include a similar, central activation, but only a lateral one

4) neither the primary somatosensory cortex nor the secondary somatosensory cortex peaked at a similar time (~55 ms & 104 ms) for the beta band









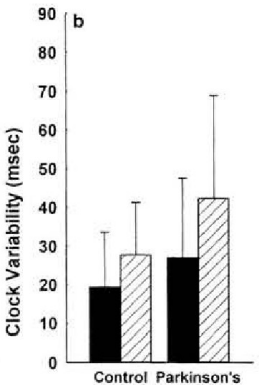
Andersen & Dalal, 2024

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INTERPRETATION

Cerebellar expectations are functionally relevant to behaviour, possibly getting translated into informed action by the thalamus

How can we learn more?
Patient groups?

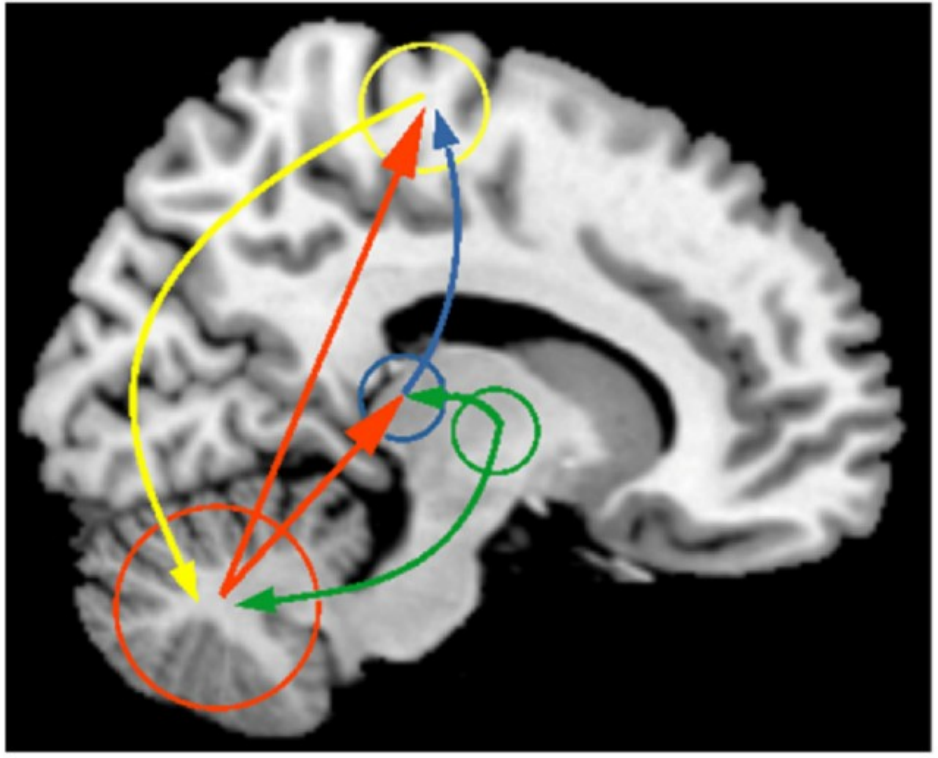


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Harrington DL, Haaland KY, Hermanowitz N (1998) Temporal processing in the basal ganglia. Neuropsychology 12:3–12. https://doi.org/10.1037/0894-4105.12.1.3

Proposed network of timing

OBJECTIVES



Current Opinion in Behavioral Sciences

(Andersen and Dalal, 2024a)

Proposed model for the cerebellum's involvement in timing action:

- *Basal ganglia passes timing information on to cerebellum*
- *Sensory expectations in the cerebellum then inform thalamus that inform primary motor cortex*
- *At the same time sensory predictions and feedback are sent back and forth between cerebellum and primary motor cortex*

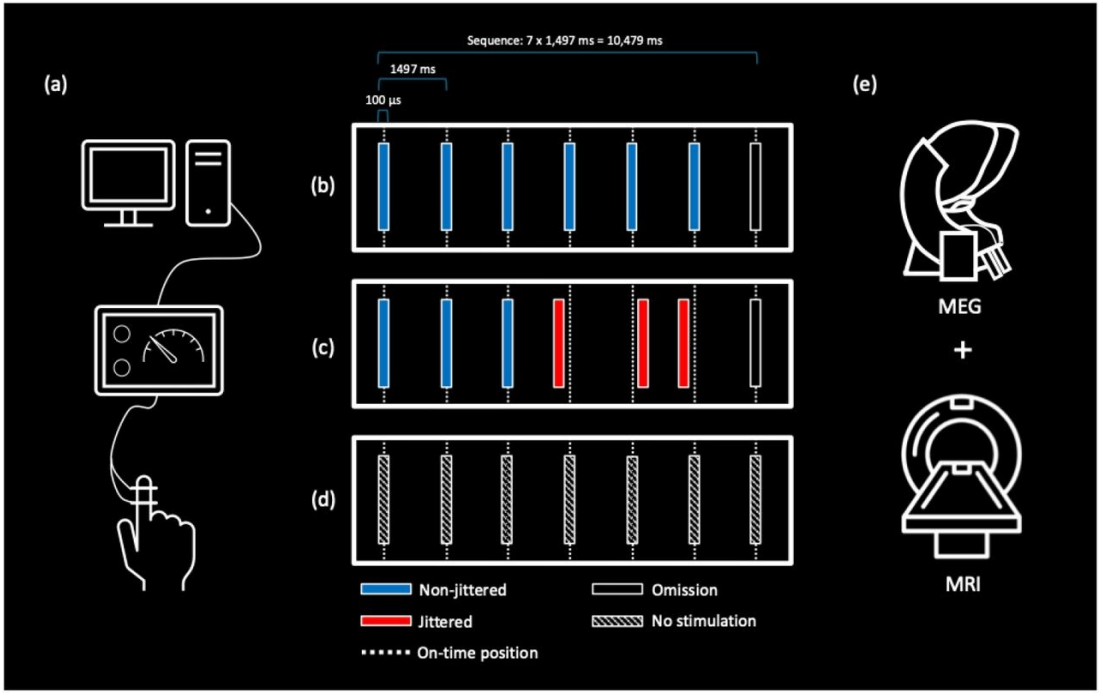
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CHALLENGES FROM AN MEG PERSPECTIVE

Thalamus and basal ganglia are deeply located with a less-than-optimal field structure

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Parkinson’s disease

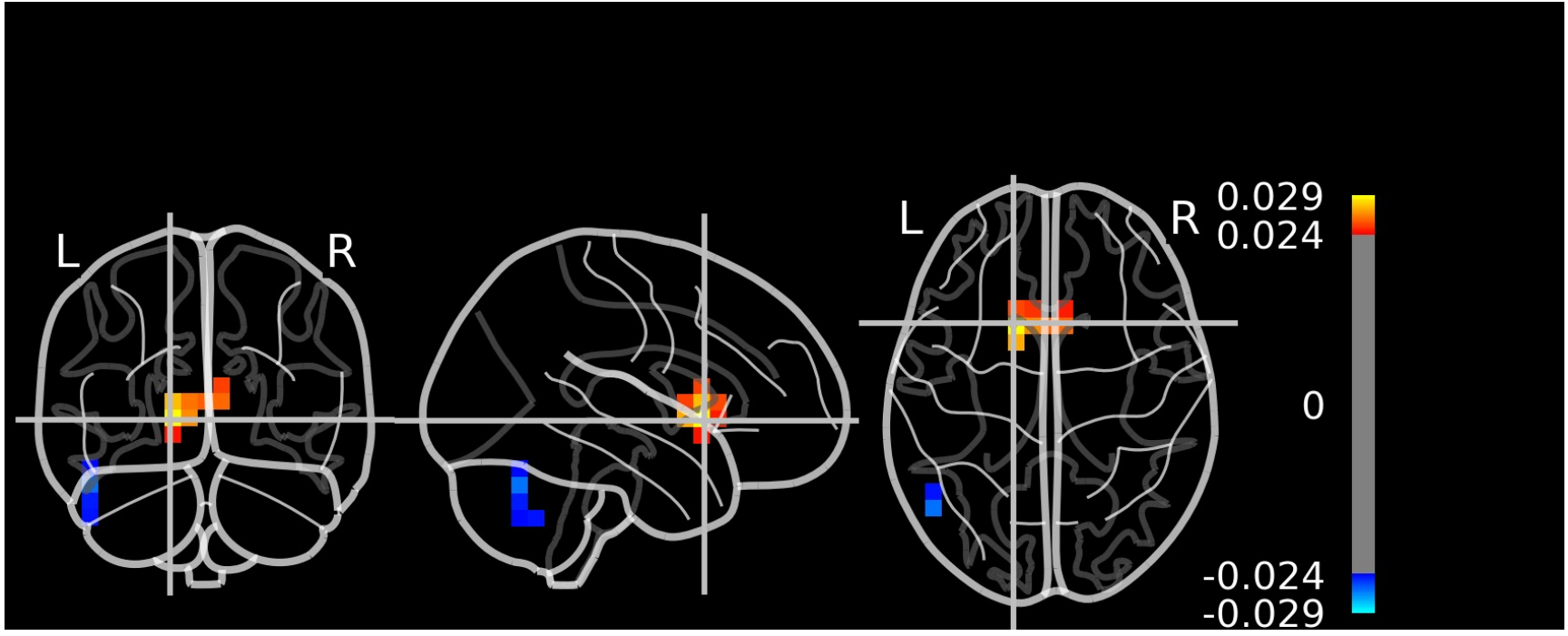


(Pando-Naude and Andersen, Accepted)

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Interaction

GROUP x REGULARITY

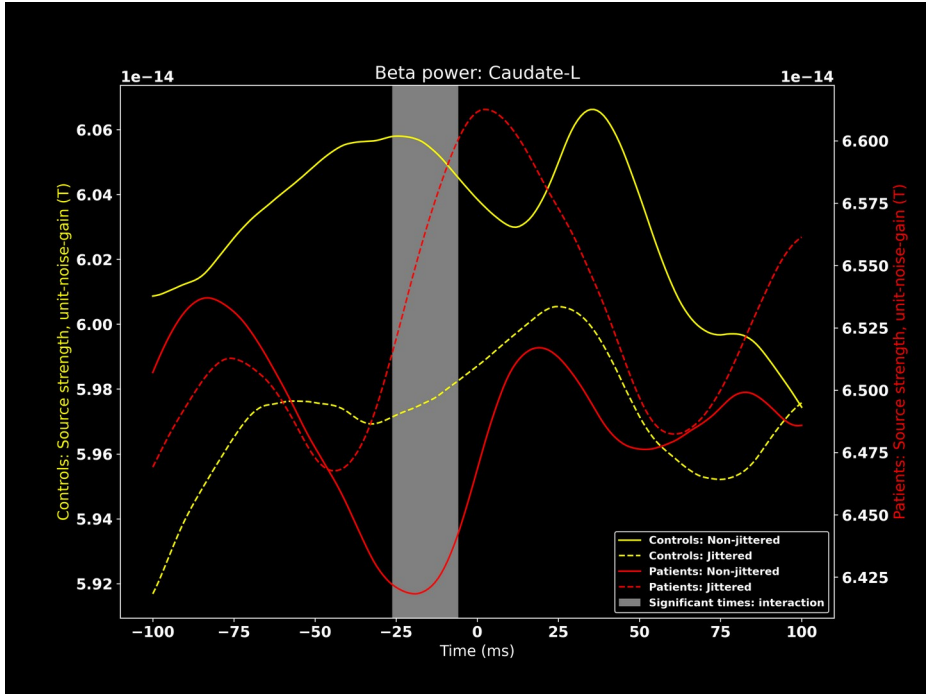


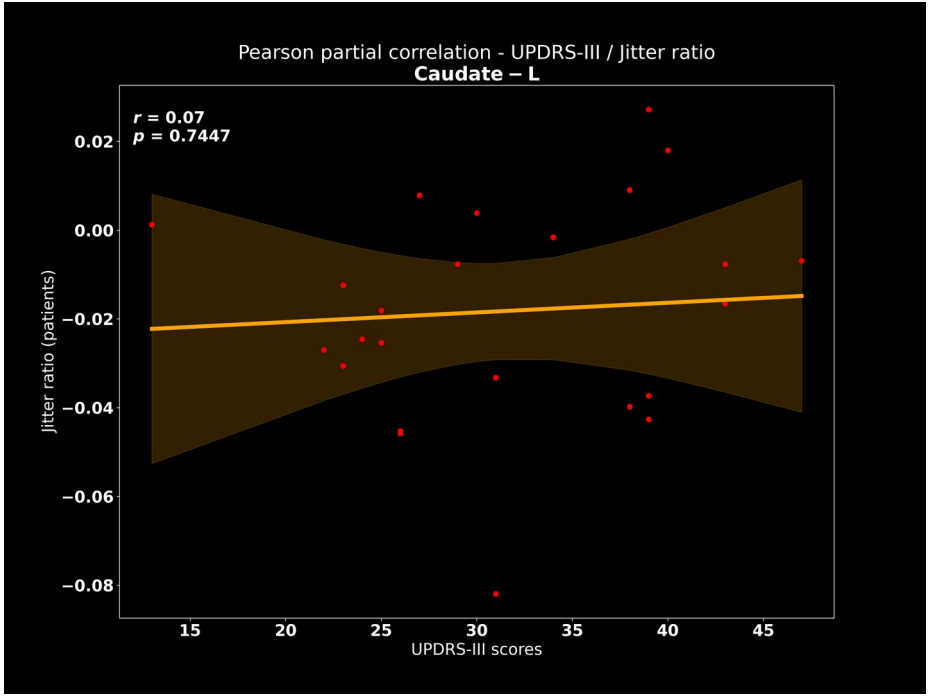
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Parkinson’s disease

CAUDATE (BASAL GANGLIA)



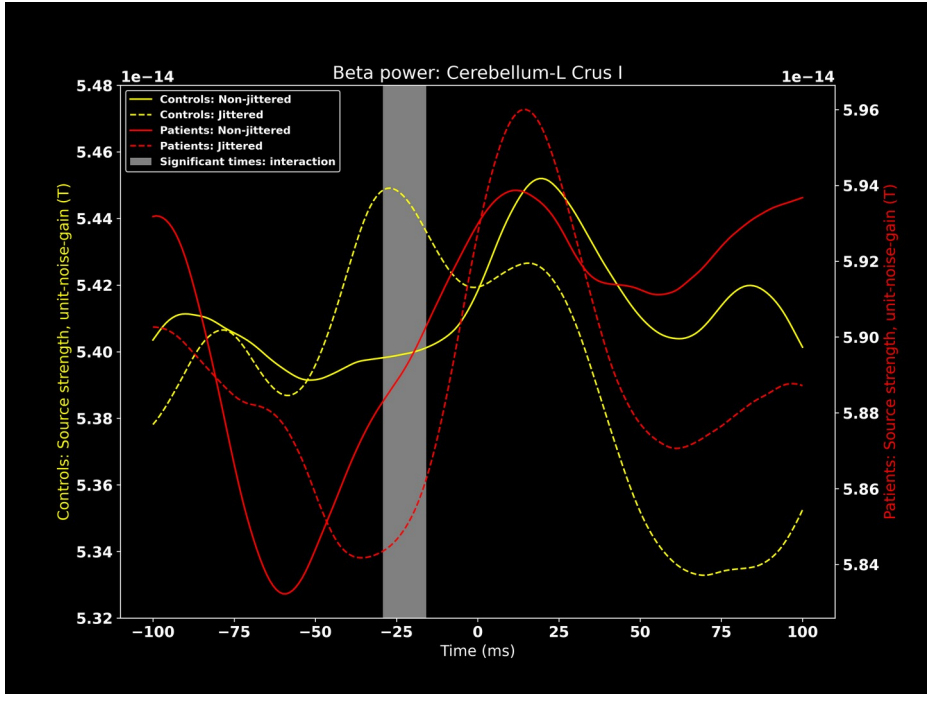


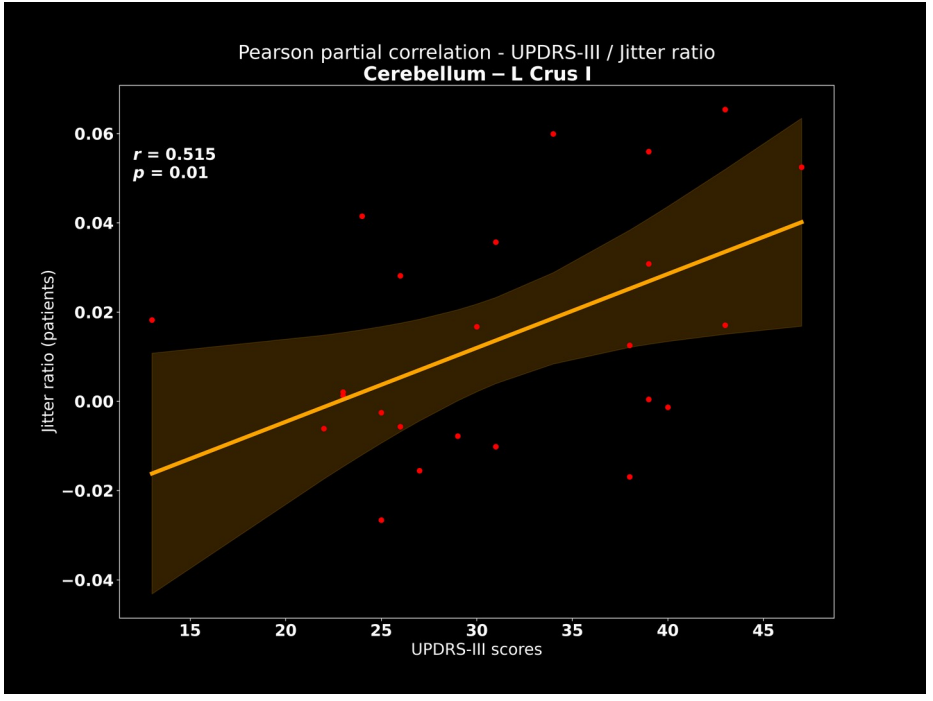
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Parkinson’s disease

CEREBELLUM

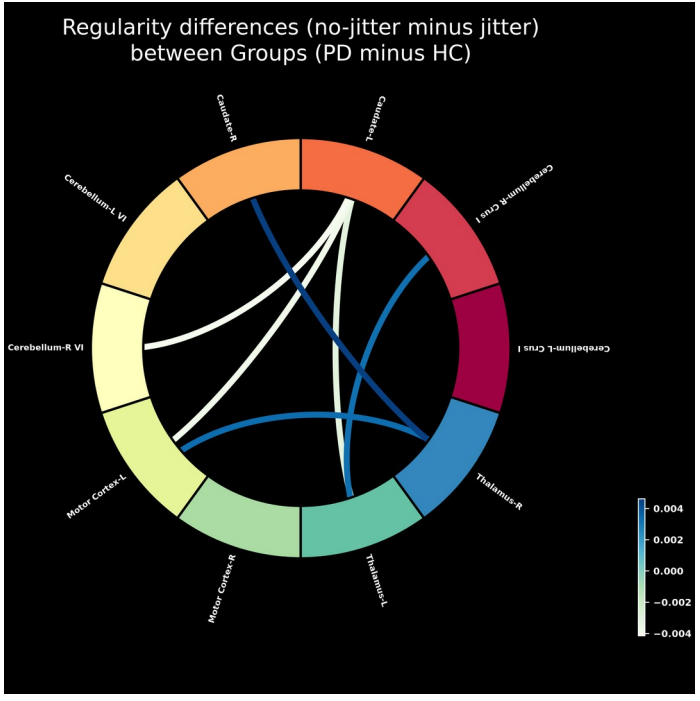


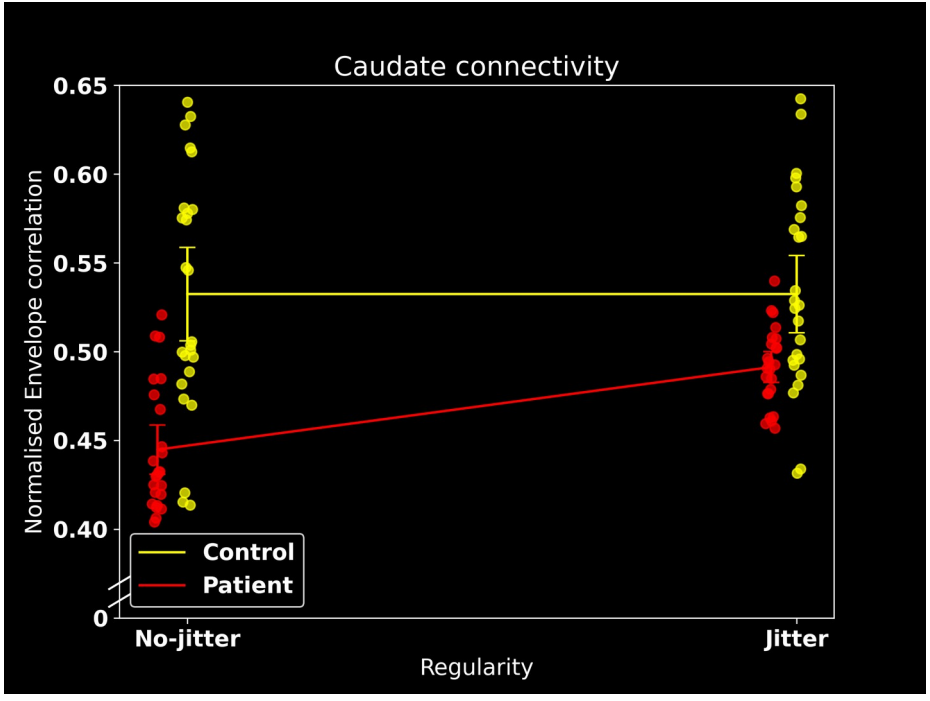


(Pando-Naude and Andersen 2025)

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Parkinson’s disease





(Pando-Naude and Andersen 2025)

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INTERPRETATION

Basal ganglia involved in timing. When dysfunctional, cerebellum may compensate.

Can we acquire causal evidence?

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An aside; without causal evidence?

Line Kruse’s presentation

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Final summary

- We can measure the cerebellum using MEG and EEG
- The cerebellum builds sensory and temporal expectations (< 4 s)
- These expectations are functionally relevant, making informed behaviour possible (through thalamus)
- Timing responses in basal ganglia and cerebellum, are different in Parkinson’s disease
- Cerebellar-basal-ganglia-thalamic disruptions in connectivity underlie differences in Parkinson’s disease patients?
- Further questions for the future:
  - Do optically pumped magnetometers allow for better cerebellar MEG? (Laura)
  - Does respiration modulate cerebellar timing-related activity? (Laura)
  - Can we causally change my proposed timing network by turning on and off DBS?

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Brief look at the future

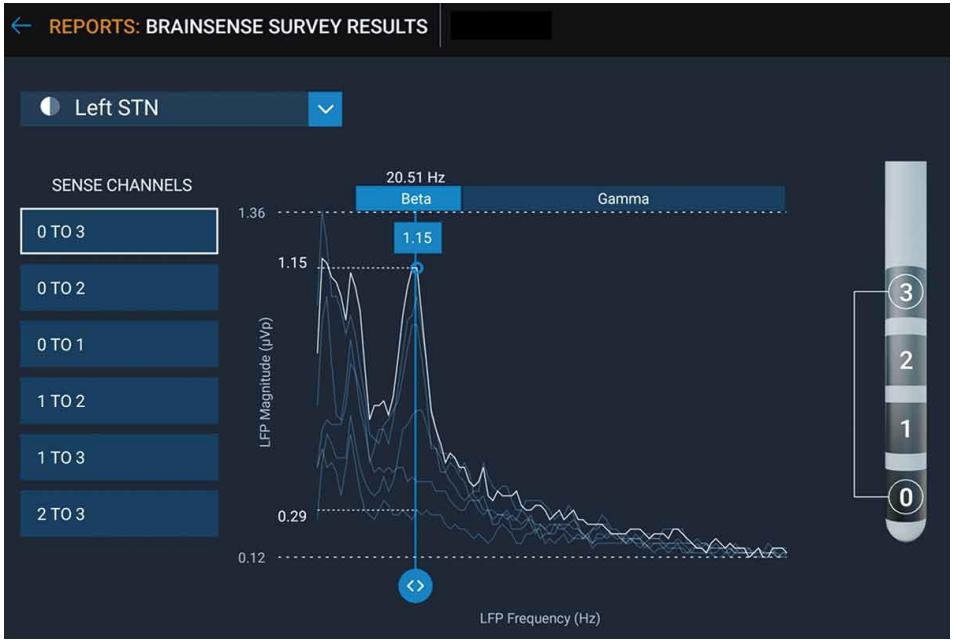
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Deep brain stimulation





By Svenskbygderna (talk) - Spiral drawing - essential tremor.jpg by Undescribed, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=78497463

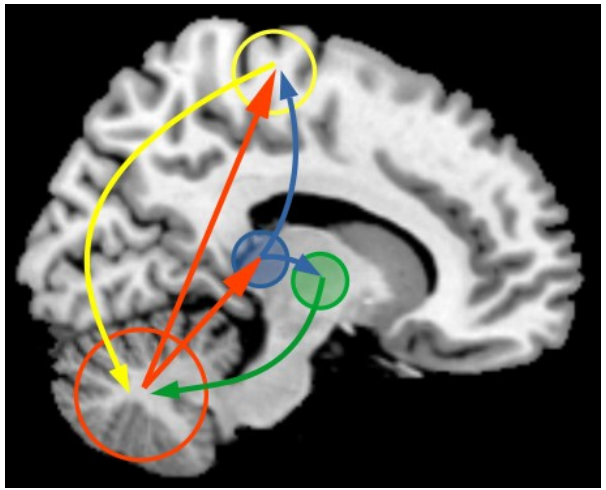


Deep brain stimulation **and** sensing

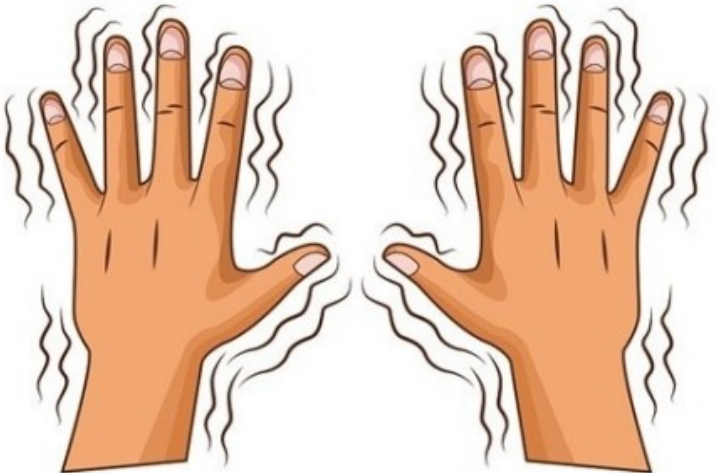
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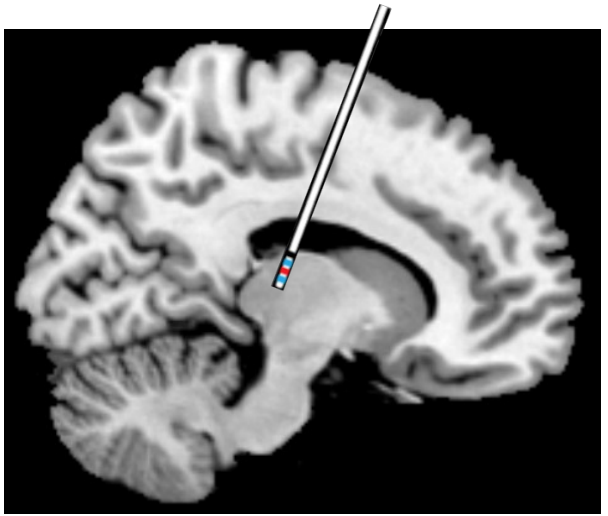
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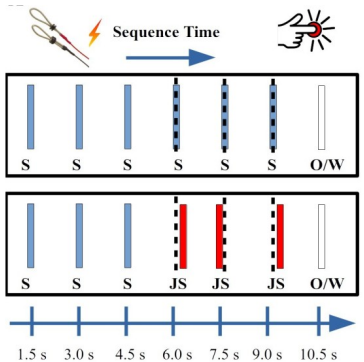
Line Elgaard Kruse Danielsen



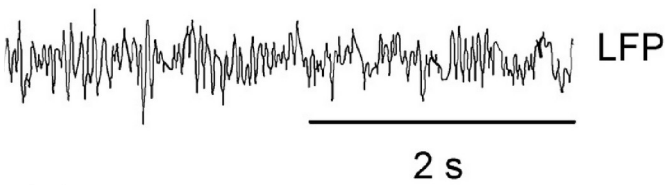
Essential tremor Parkinson's disease











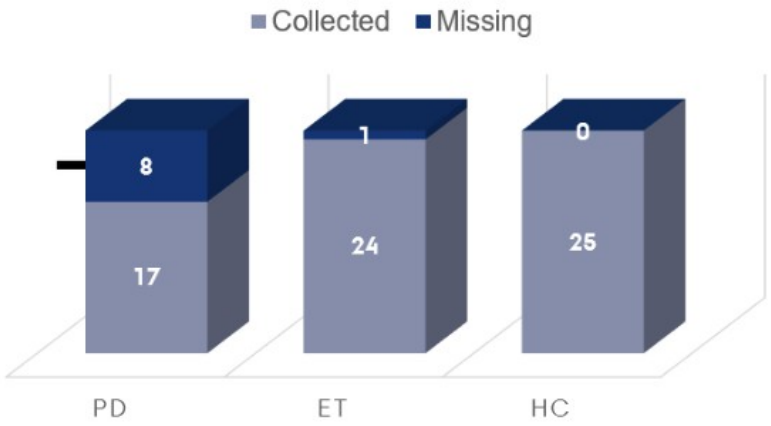
The cerebellum translates sensory expectations into informed action

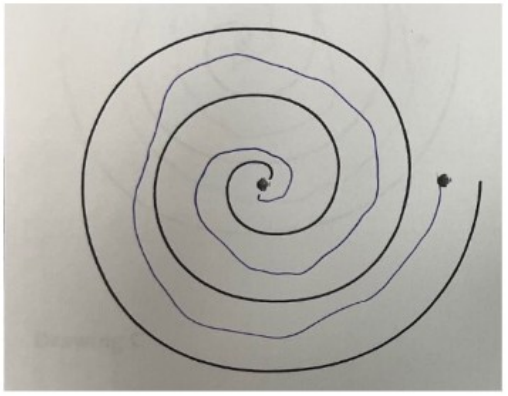
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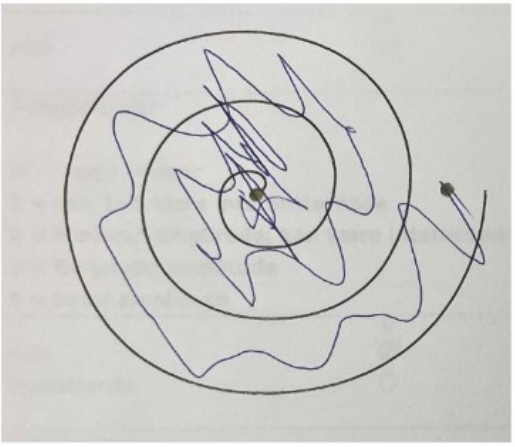
CFIN www.cfin.au.dk

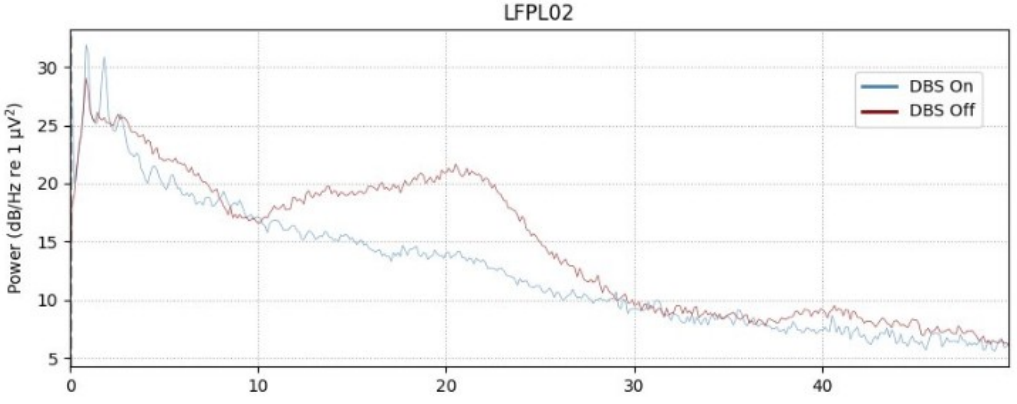
AARHUS UNIVERSITY HOSPITAL

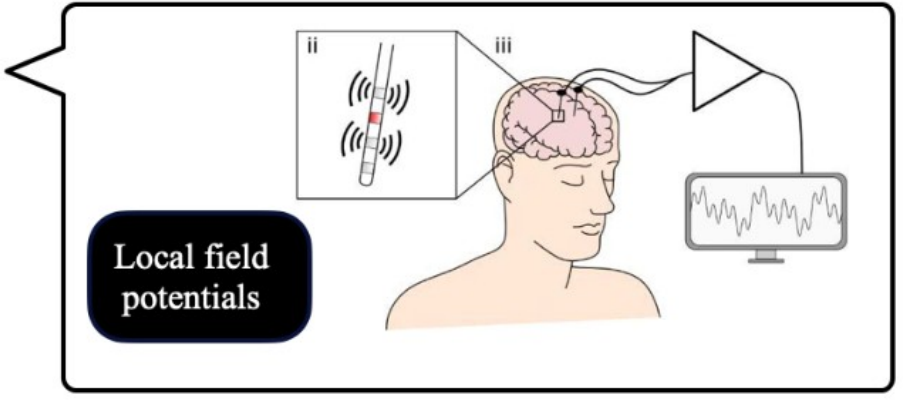
DATA COLLECTION STATUS









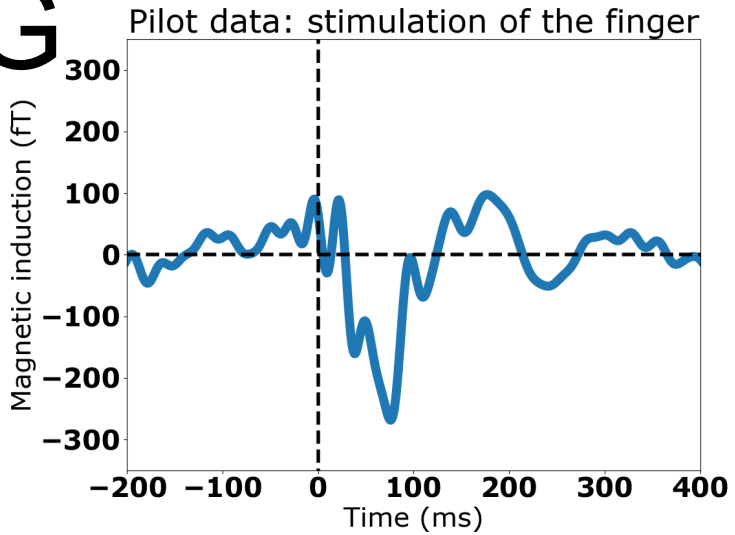


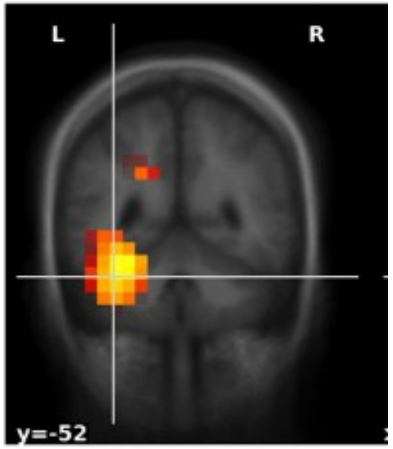
OPM MEG



Laura Bock Paulsen
PhD student





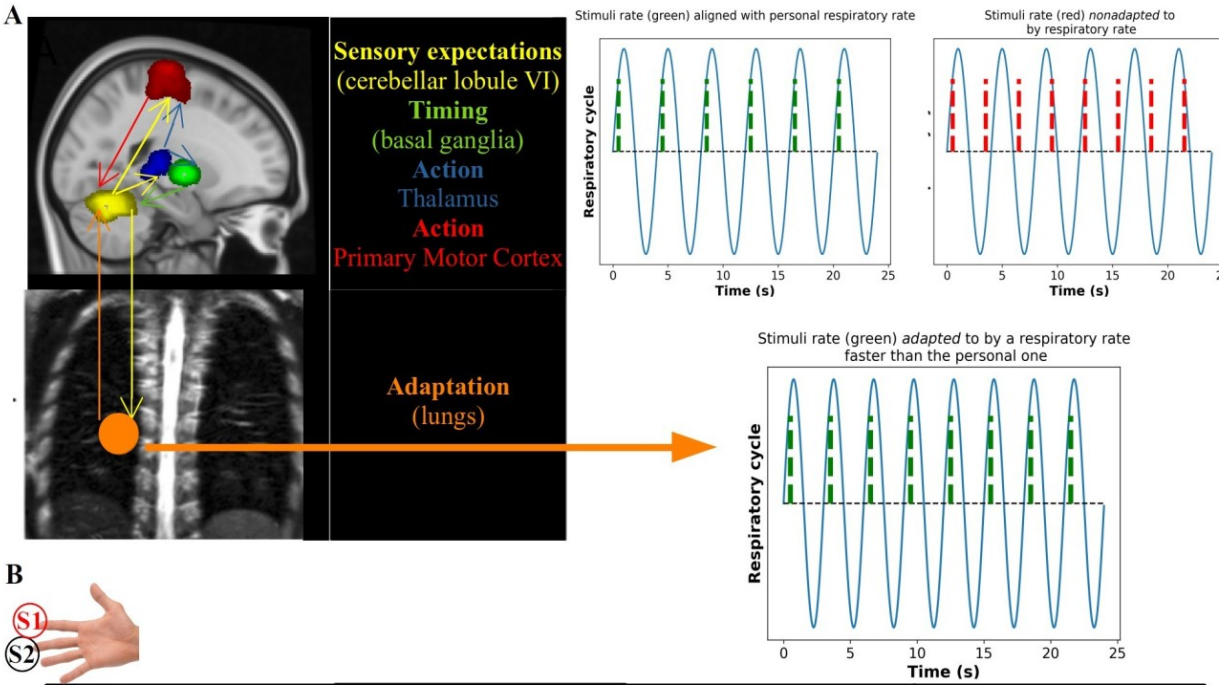


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Respiration and the cerebellum



Laura Bock Paulsen
PhD student





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The course plan

Week 37:
Monday: Workshop paradigm: Measuring visual subjective experience
MR Recordings (Lau)

Week 38:
Monday: What is it all about? (Lau)
Workshop 1: MR-recordings (Laura & Lau)

Week 39:
Monday: Basic physiology and Evoked responses (Lau)
Wednesday: Preprocessing and evoked responses (Laura)

Week 40:
Monday: Forward modelling and inverse modelling (Lau)
Wednesday: Source reconstruction (Laura)

Week 41:
Monday: Multivariate statistics (Laura) (Lecture)
Wednesday: Multivariate statistics (Laura) (Coding)
Deadline for feedback: Video Explainer

Week 43:
Monday: Inverse modelling: beamforming (Lau) (Lecture)
Wednesday: Inverse modelling: beamforming (Laura) (Coding)

Week 44:
Monday From single subject analysis to group analysis (Laura)
Wednesday: Work on your reports (Laura)

Week 45:
Monday: What about that other cortex? - the cerebellar one (Lau)
Wednesday: Work on your reports, continued (Laura)

Week 46:
Wednesday: Optically pumped magnetoencephalography (Laura)
Wednesday: Respiration (Laura)
Deadline for feedback: Lab report

Week 47:
No teaching: prepare presentations!

Week 48:
Monday: Oral presentations (part 1) (Lau)
Wednesday: Oral presentations (part 2) (Lau)

Week 49:
Monday: Oral presentations (part 3) (Lau)
Wednesday: Oral presentations (part 4) (Lau)

Week 50:
Wednesday: What was it all about (Lau)

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